

Experiment protocol template

実験計画のテンプレート



English:

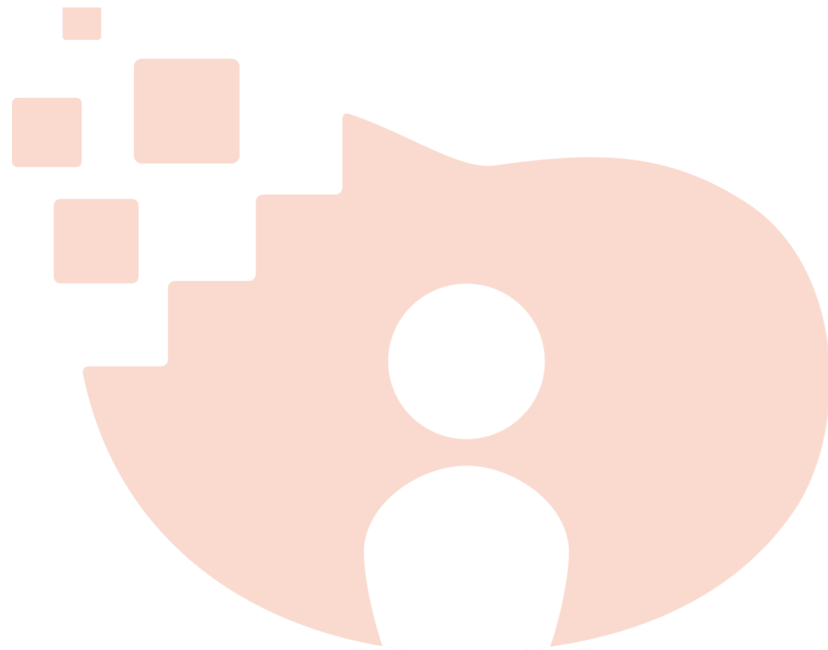
https://docs.google.com/document/d/118Crarfw62Z1odOkCYw2bjbrqtM4fQ_NtFUstNjxic/edit?usp=sharing



Japanese:

<https://docs.google.com/document/d/1RrFIGOOfHreyHFAYICkkuOFIdev8qmTUfftALoyevYg/edit?usp=sharing>

Designing **User** Studies for XR Research



Monica Perusquia-Hernandez

Cybernetics & Reality Engineering Lab

Division of Information Science

NAIST

Contents

もくじ

- Why do CS need experiments with people?
- Reproducibility crisis and Open Science
- Experiment protocols and Experiment design
- Power analysis
- Data analysis using R
- Ethical approval

Contents

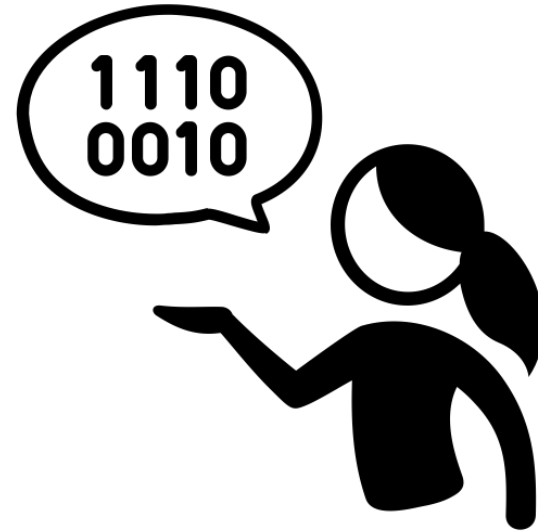
もくじ

- **Why do CS need experiments with people?**
- Reproducibility crisis and Open Science
- Experiment protocols and Experiment design
- Power analysis
- Data analysis using R
- Ethical approval

I am a computer scientist



Created by N.K.Narasimhan
from Noun Project



Created by Mazil
from Noun Project

I am a computer scientist

**Why do I need to care
about people?**

Interfaces are ubiquitous

As Computer Scientists and Engineers, we build many

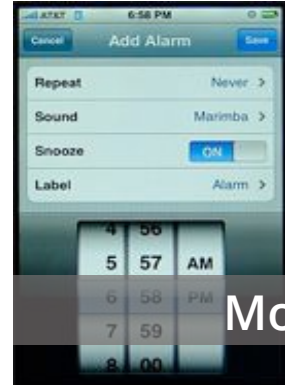
Interface

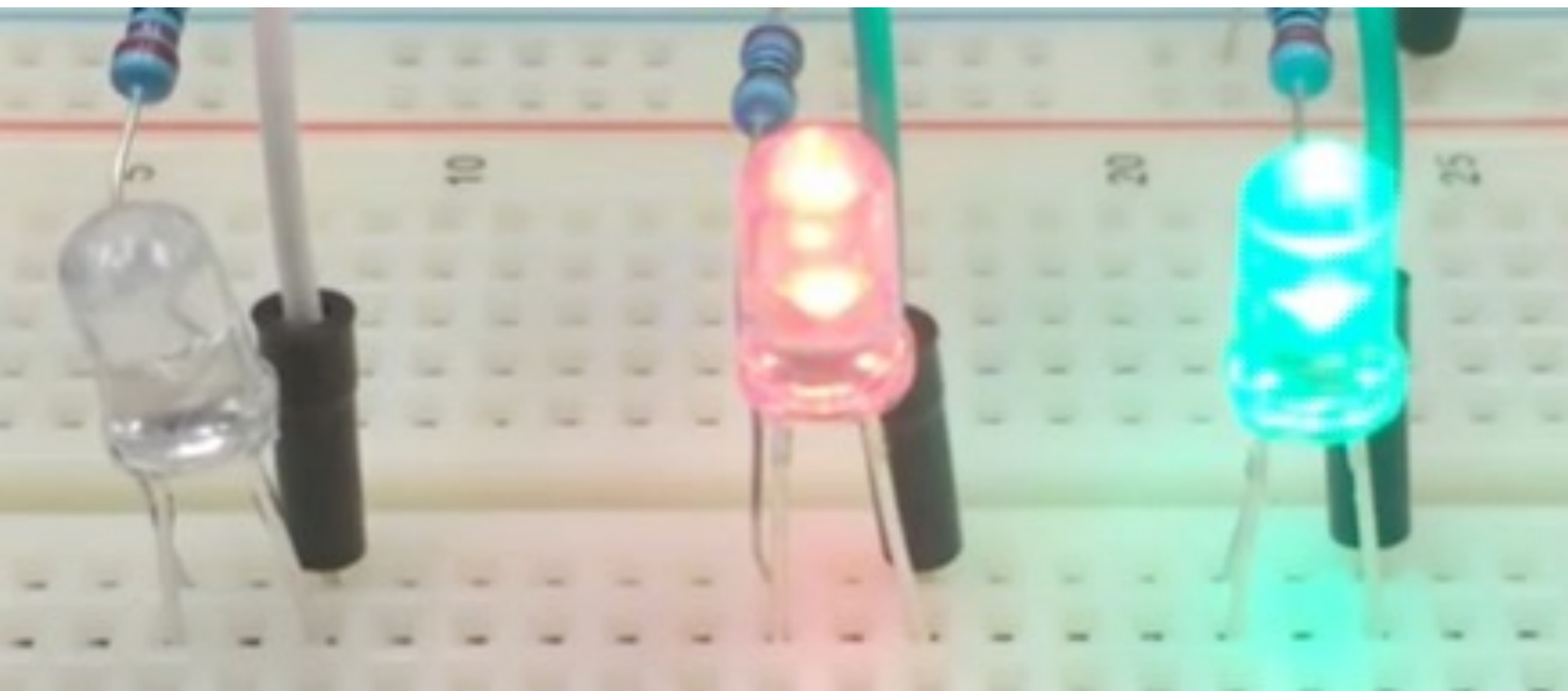
An interface is a device or a system that unrelated entities use to interact.

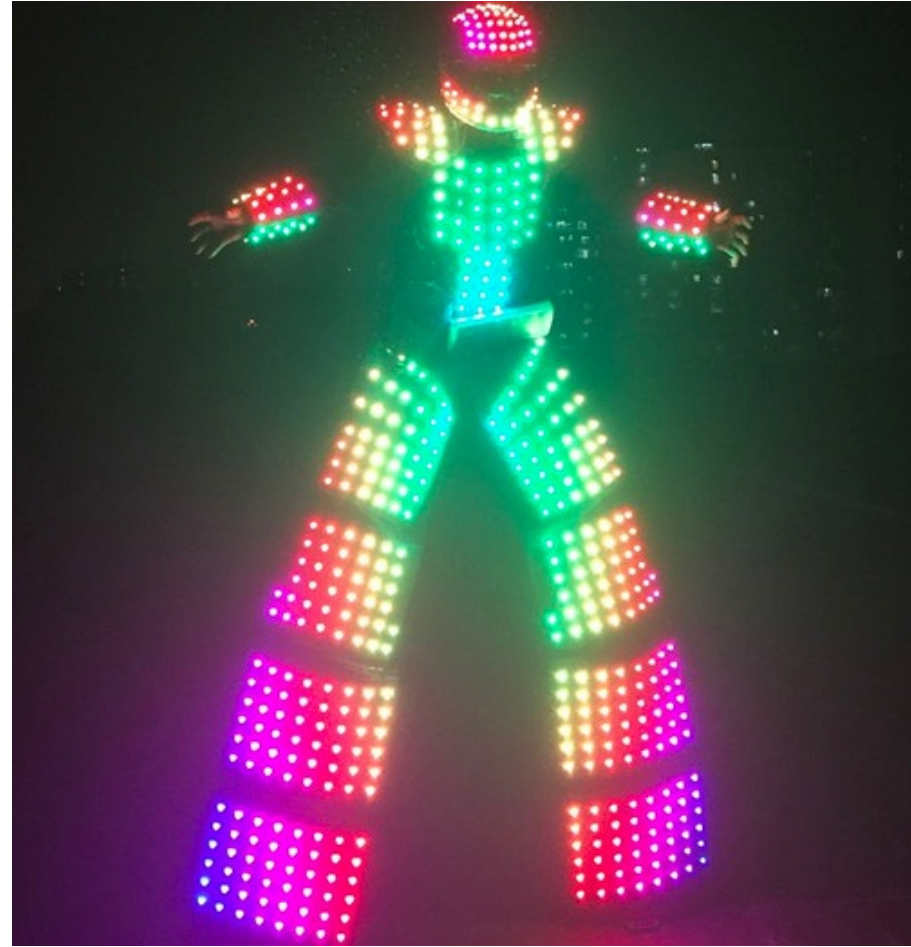


Computer Interface

The meeting point between a computer and something or someone outside of it.







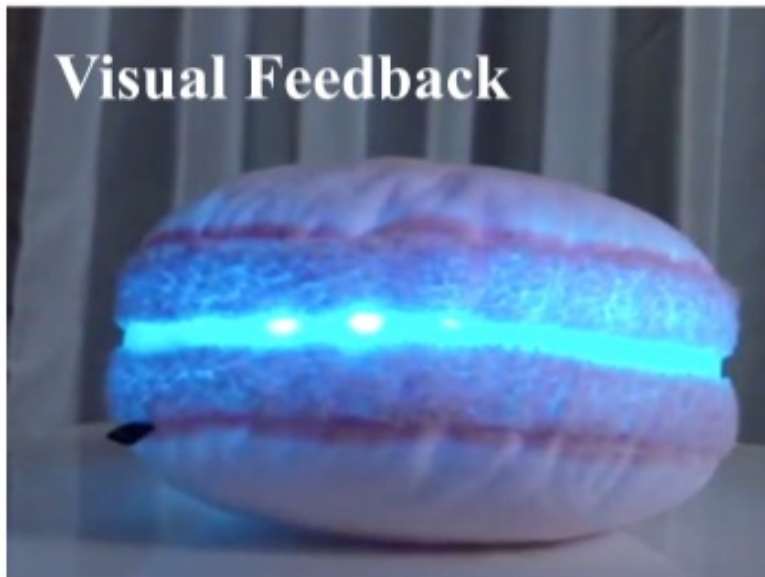
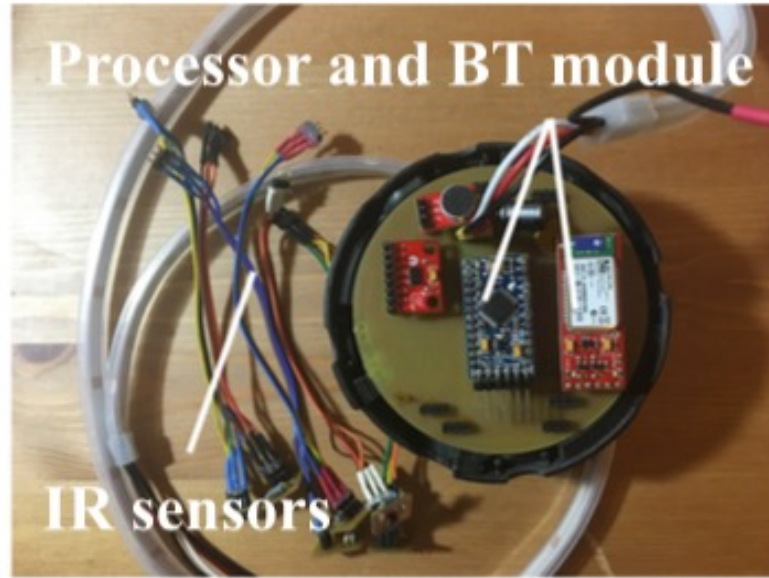
We (engineers) love LEDs!

Network Switch Port LED and Status LED Guidelines

Blinking amber implies a blink rate of 33ms. *Slow blinking amber* indicates a blink rate of 500ms, with a 50% on/off duty cycle. In other words, a slow blinking amber LED is amber for 500 ms and then off for 500ms.

Activity	Max Speed indication	Lower Speed Indication
Physical Link Down	Off	Off
Physical Link UP	Solid Green	Solid Amber
Link Tx/Rx Activity	Blinking Green	Blinking Amber
Beaconing	Slow Blinking Amber	Slow Blinking Amber
Fault	Slow Blinking Amber	Slow Blinking Amber

**What does the slow
blinking amber mean?**

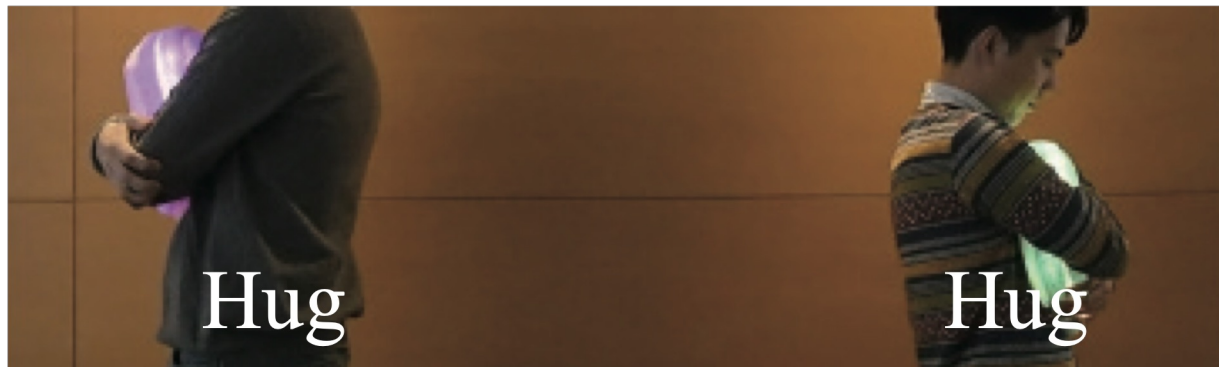
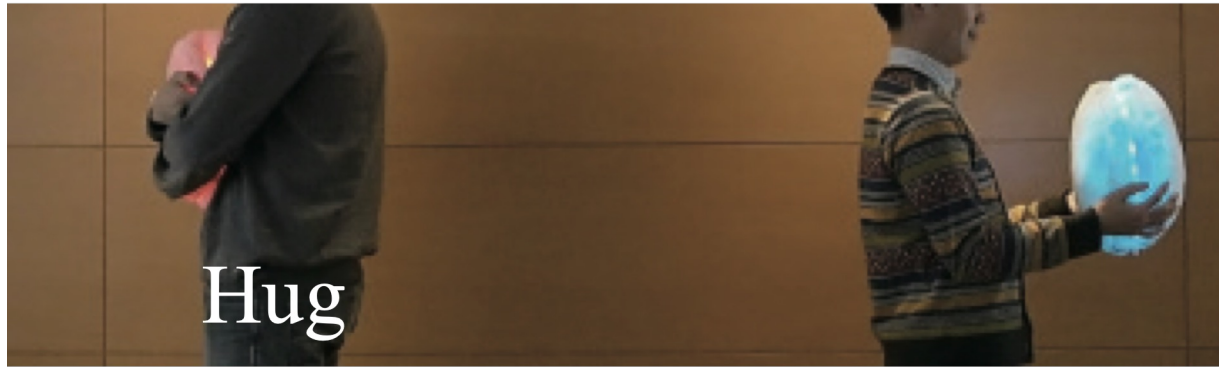


Awesome
piece of
technology

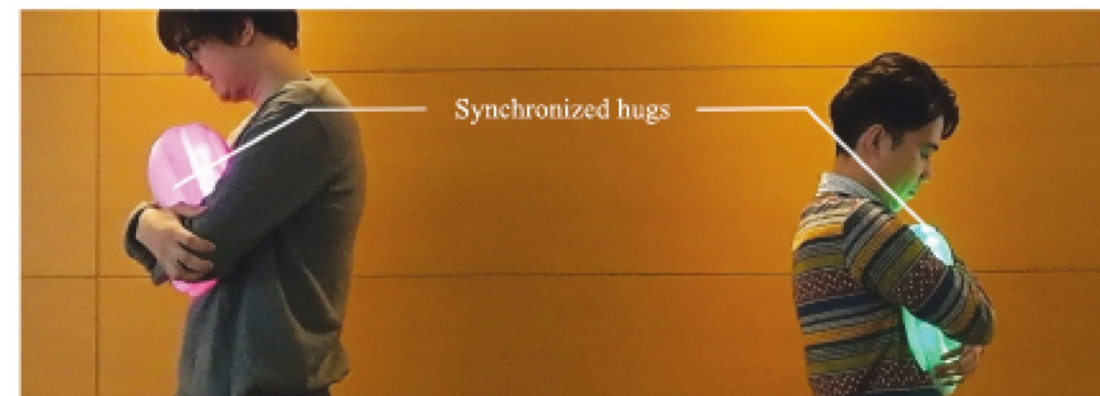
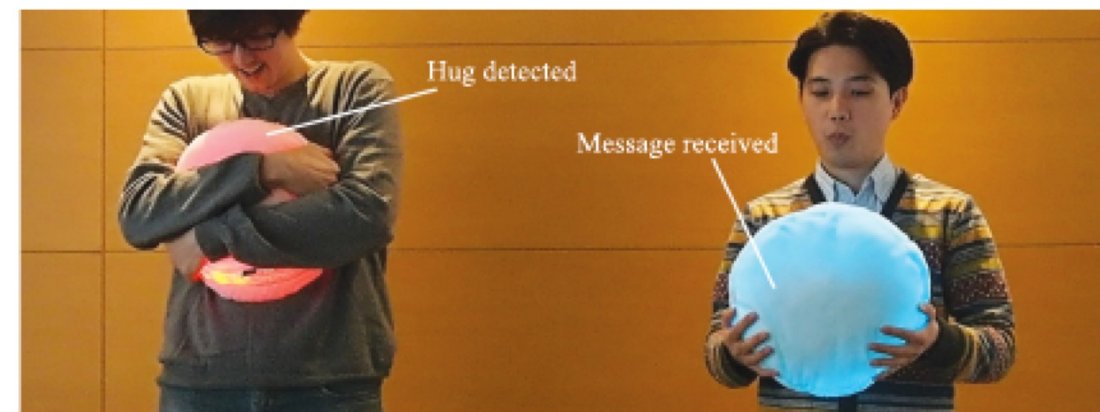
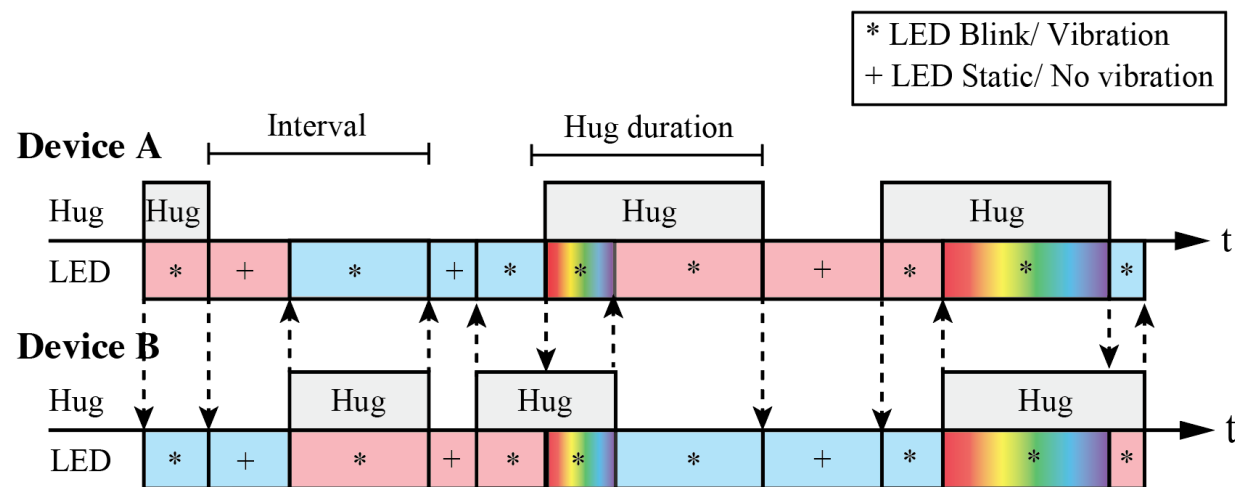
Soft sensor

It **affords**
hugging

Fancy LED interface



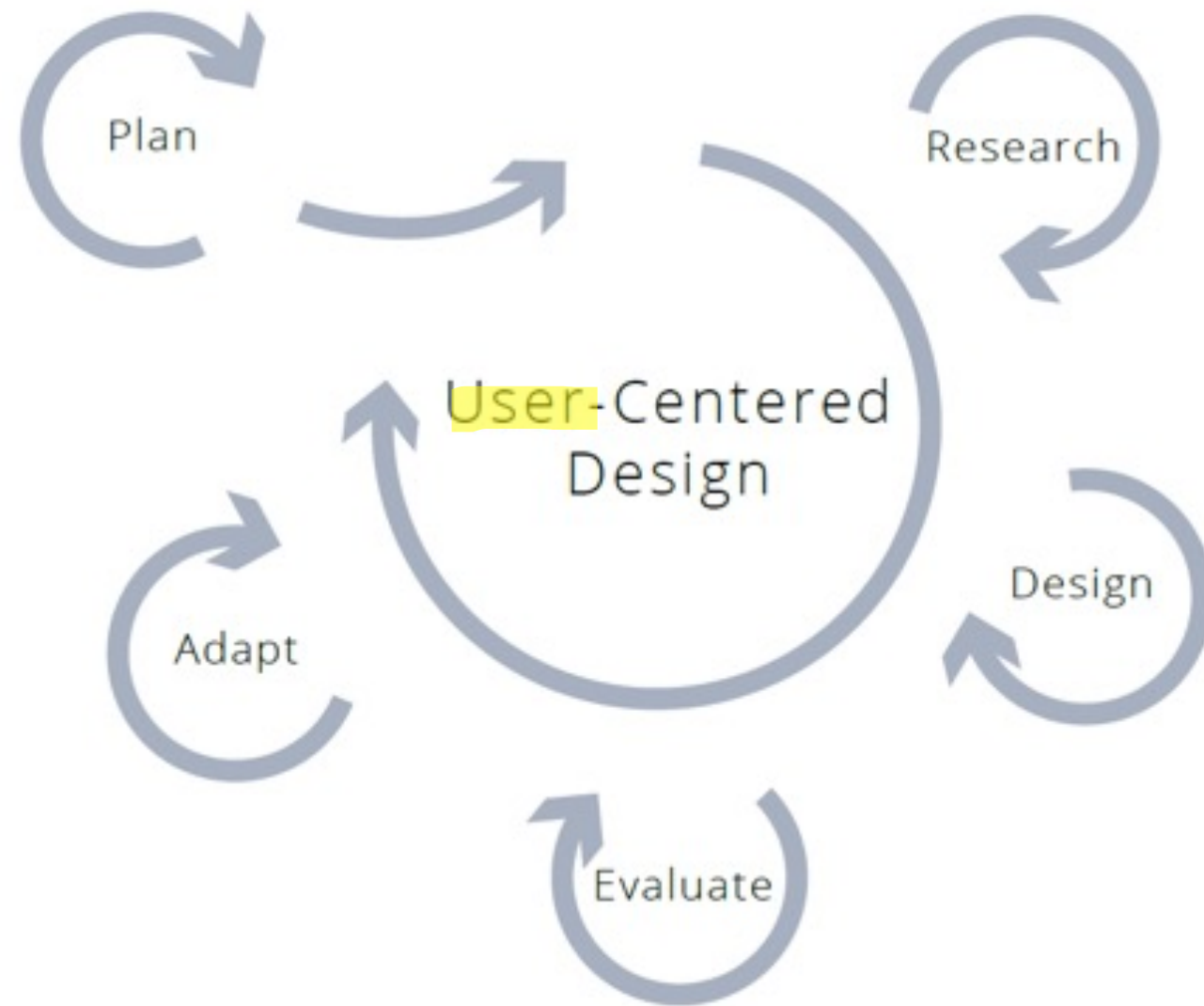
Meant for social communication



Aimed for subtle feedback, but it is very difficult to understand!

**A User Interface
is like a joke**

**if you have to
explain it
is not that good**

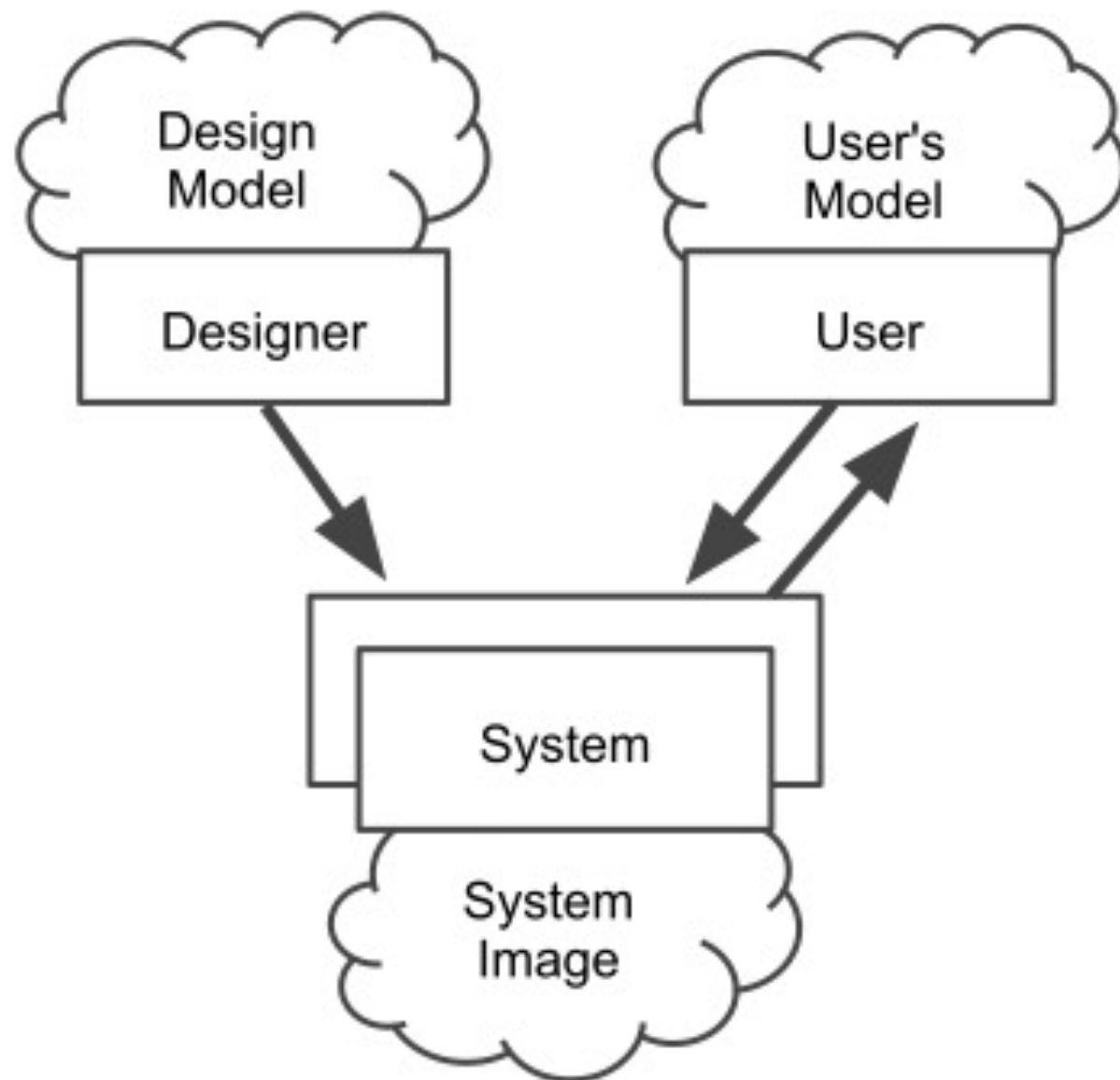




An interface is a communication tool.

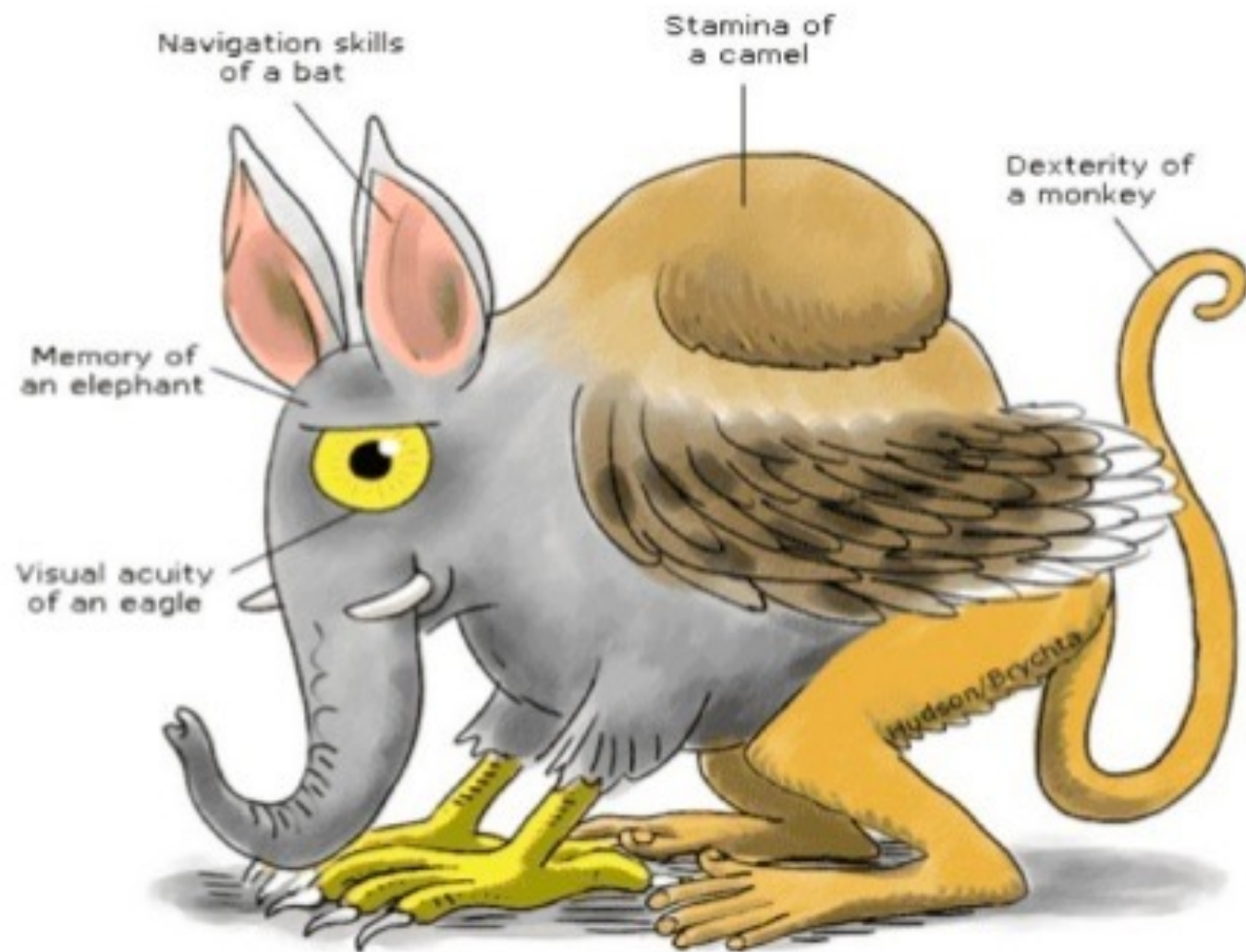
As with any other language, we have to agree on its meaning.

What people understand depends on the **mental models** they have.



You are NOT the user

“The perfect user”



Copyright © 2001-2004 Syntagm Ltd

Why is hard to predict how users will react?

People are not rational

People are not like you

People will not use your product the way you imagine

Even people themselves cannot tell what they will like

This is why **psychology-based design guidelines** are so
useful

REVISED & EXPANDED EDITION

The DESIGN
of EVERYDAY
THINGS

DON
NORMAN



Design principles

Visibility	Can I see it?
Feedback	What is it doing?
Affordance	How do I use it?
Mapping	Where I am and where can I go?
Constraint	Why can I do that?
Consistency	I think I have seen this before...

Design principles

Visibility	Can I see it?
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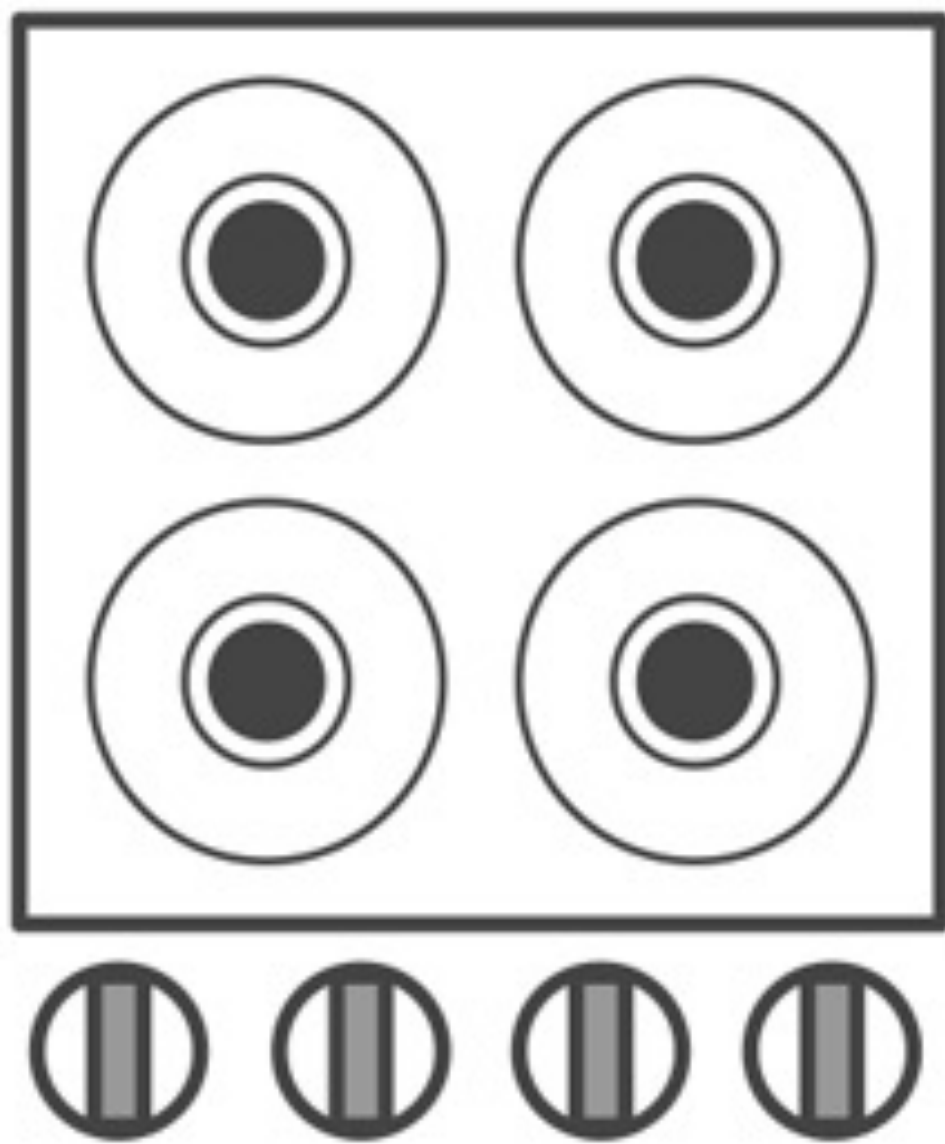
Affordance

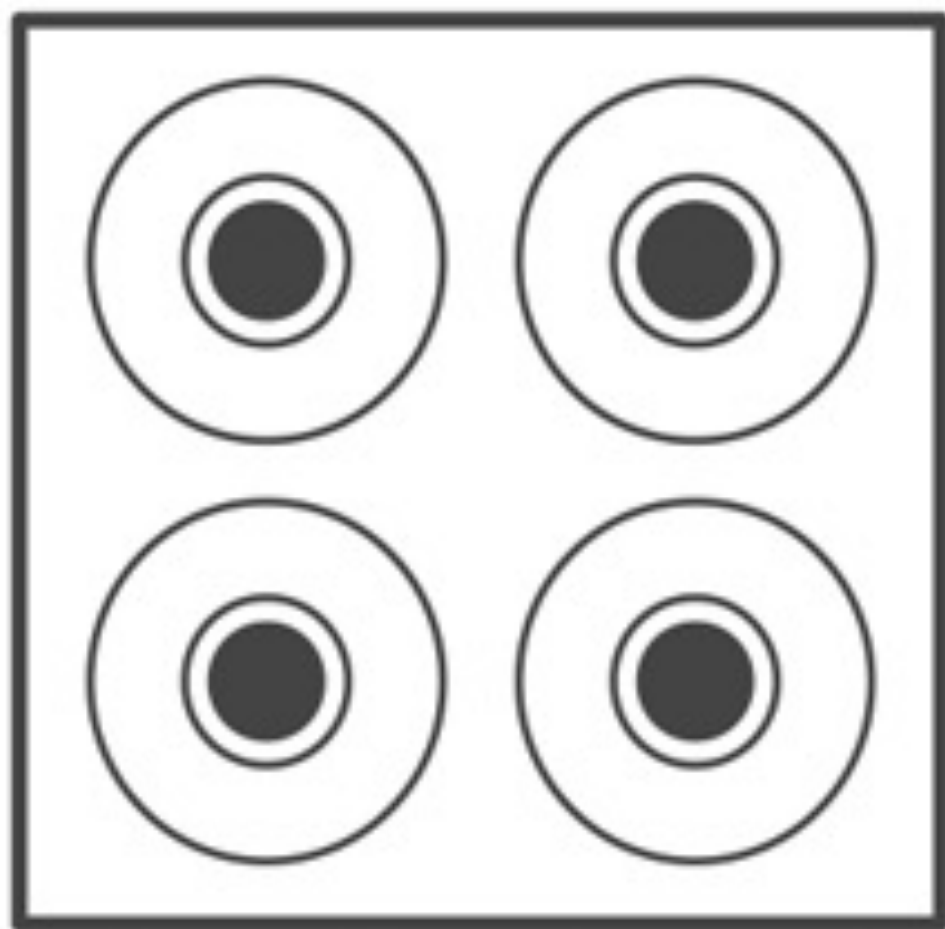
Perceived and actual properties of an object that give clues to its operation











Back
Right



Front
Left



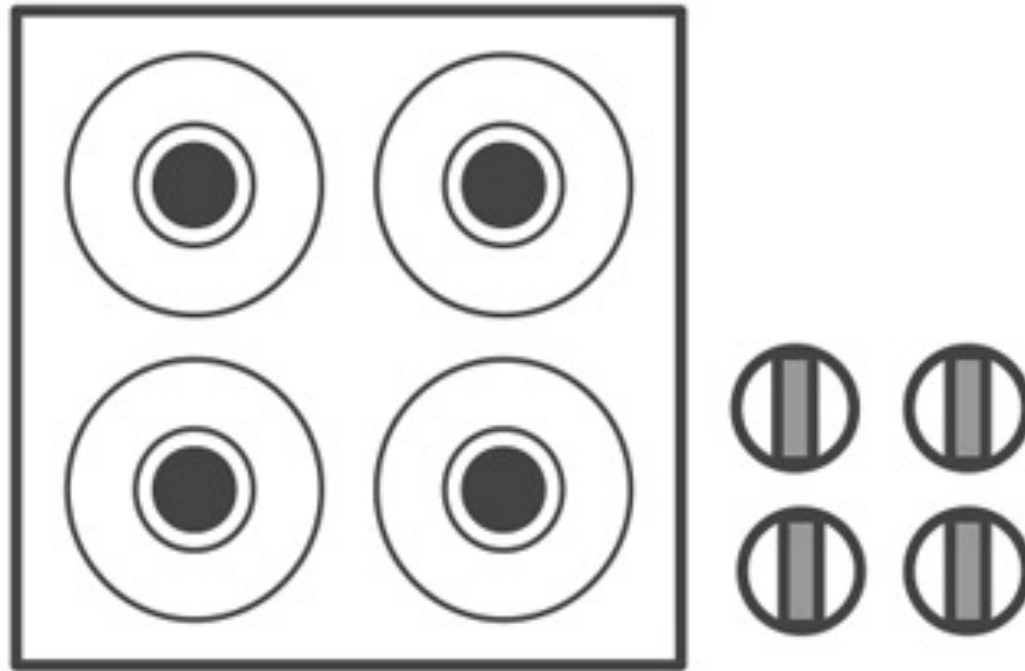
Back
Left

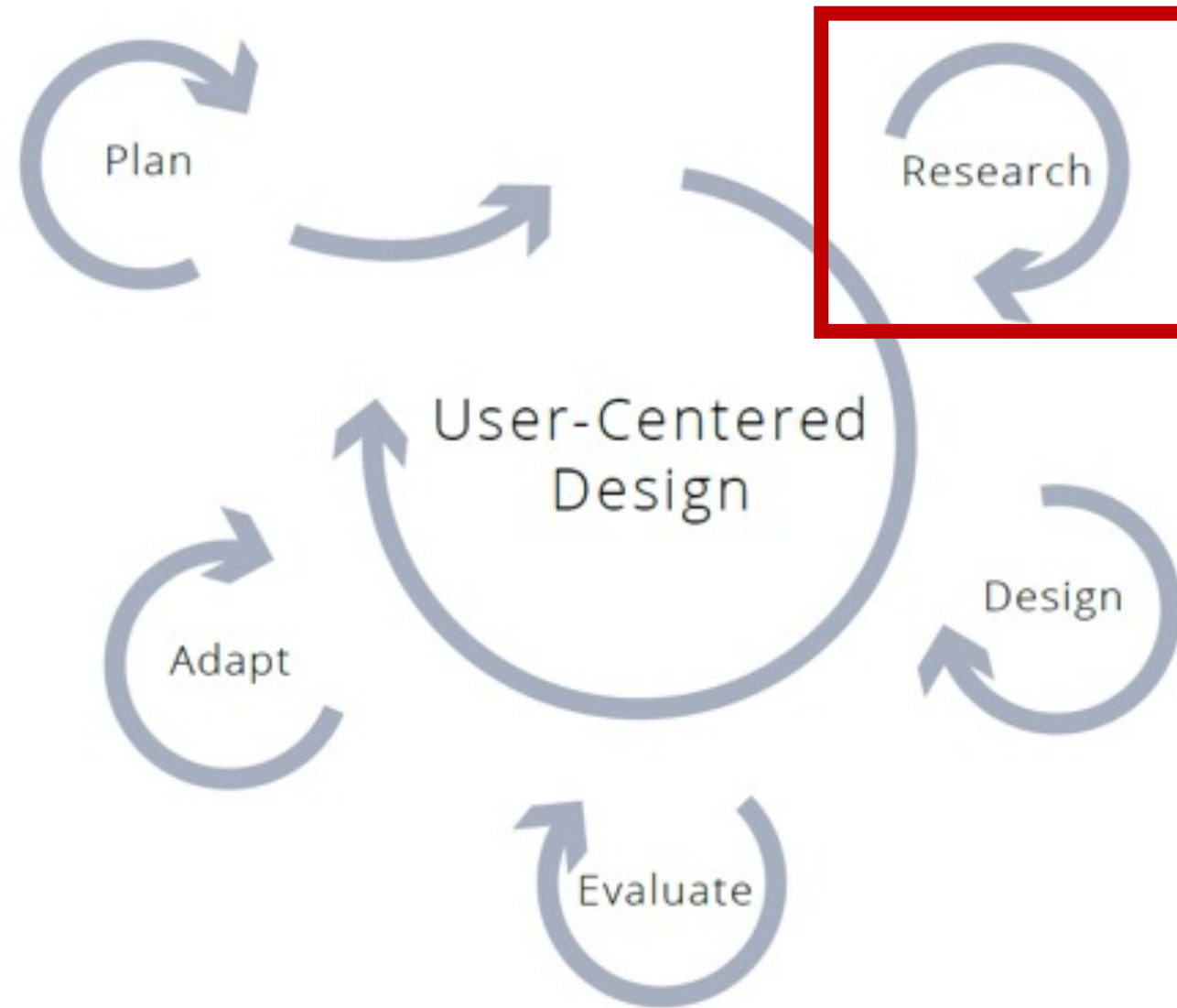


Front
Right

Natural Mapping

The relationship between a control and its resulting function





Guidelines are helpful

But **people** will always surprise you!



YOU HAVE CHRONIC
MAHJOBBISS CRAPPUS
BUT THAT'S NOT WHY
YOU PUKED.



HAVE YOU BEEN
EXPOSED TO ANY
USER INTERFACES
DESIGNED BY
ENGINEERS?



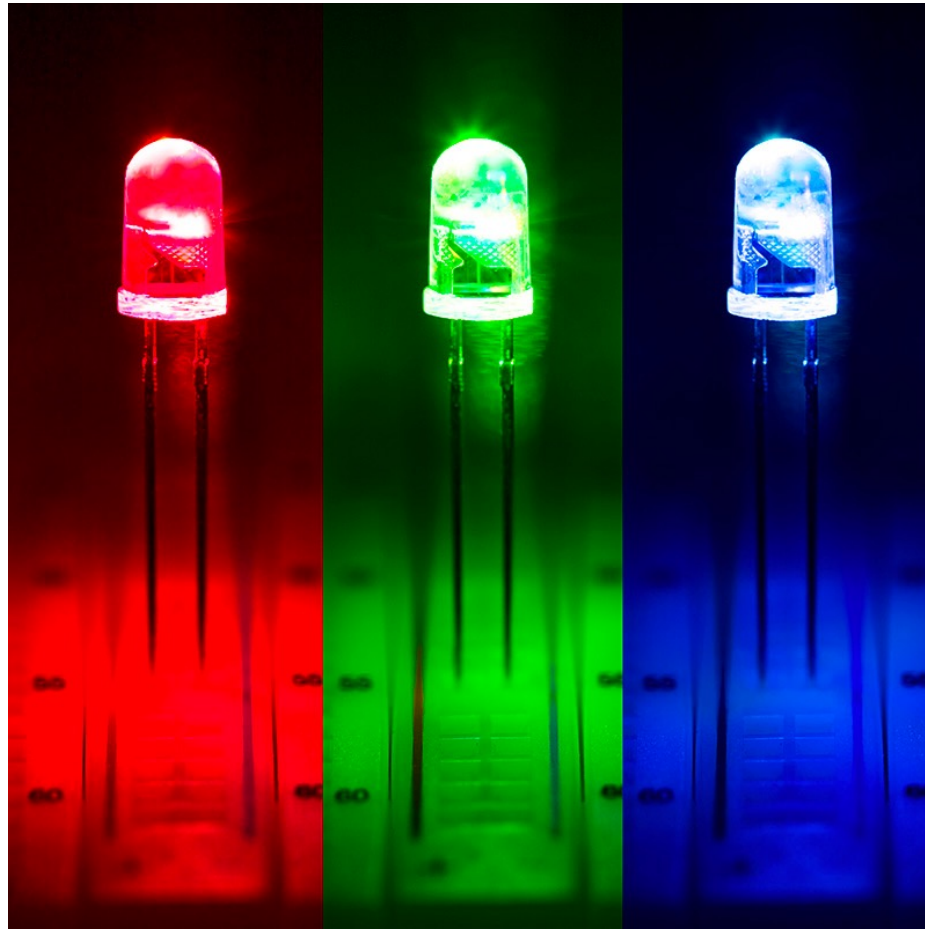
YES.

YOU HAVE INTERFACE
POISONING. YOU'LL
BE DEAD IN A WEEK.

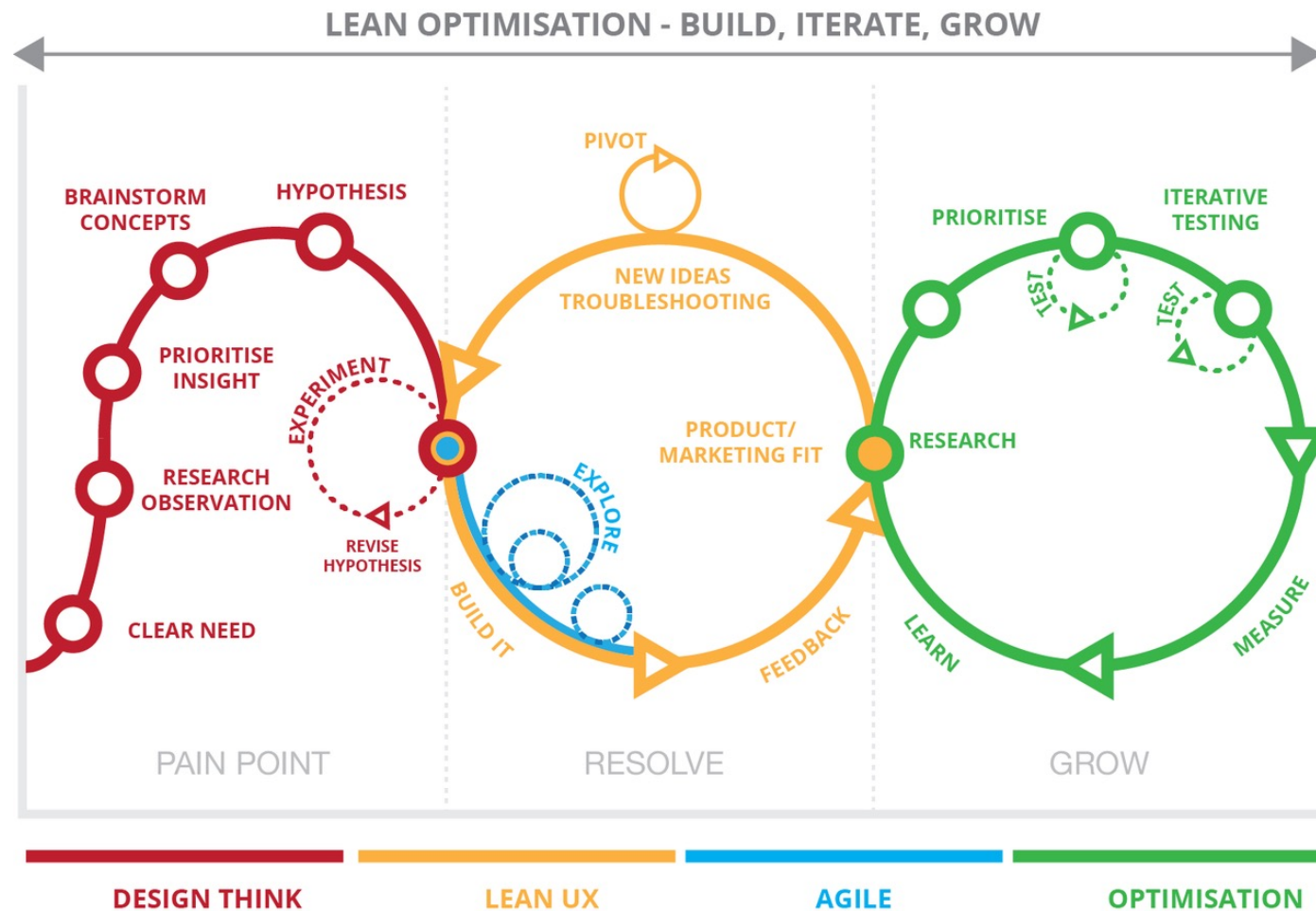


www.dilbert.com

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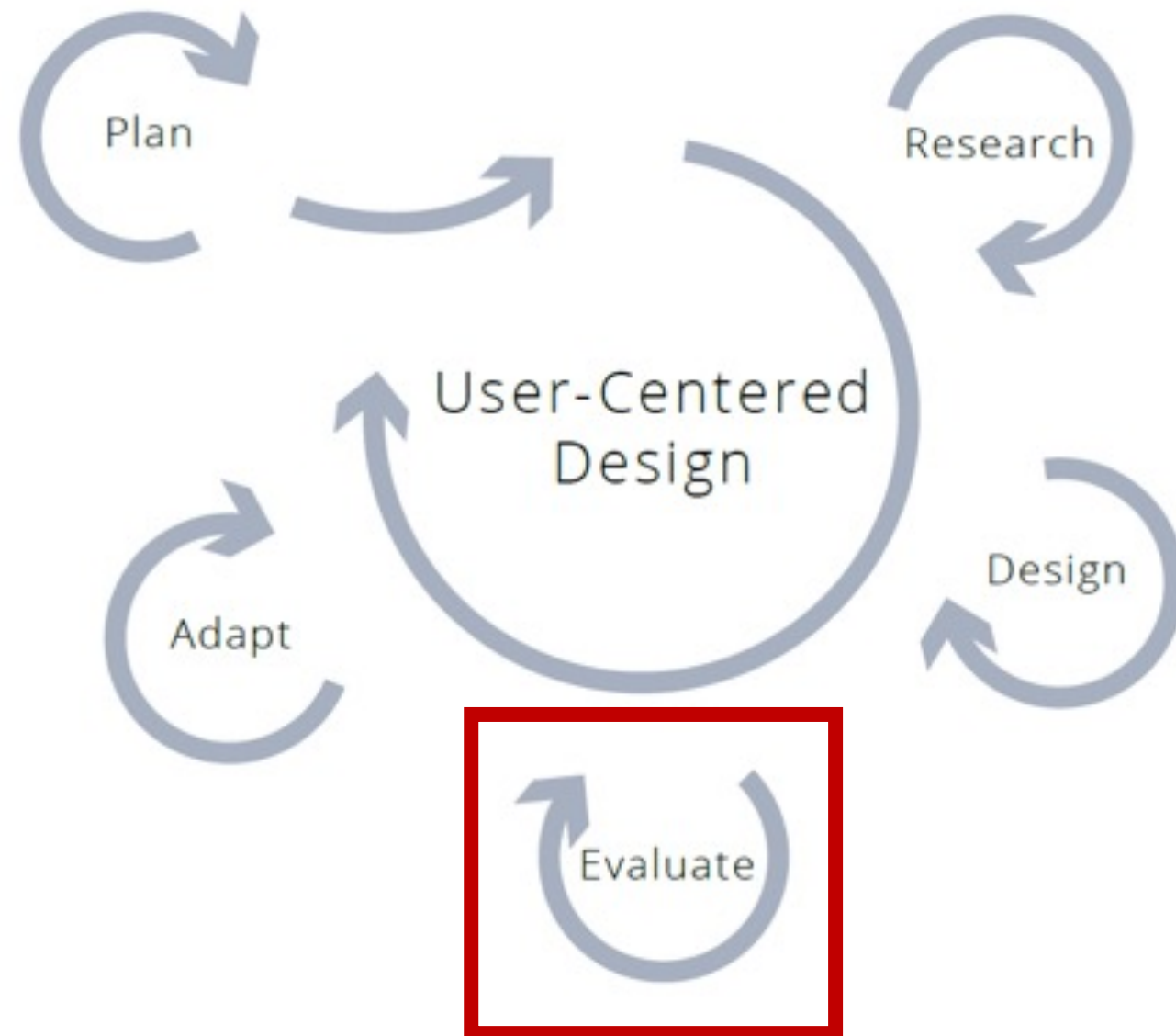


User-centered design is better
than any engineering logic



Ask the user, then ask again

Fail fast, recover fast



Kamebō: Personal robot assistant for elderly with dementia



I am a computer scientist

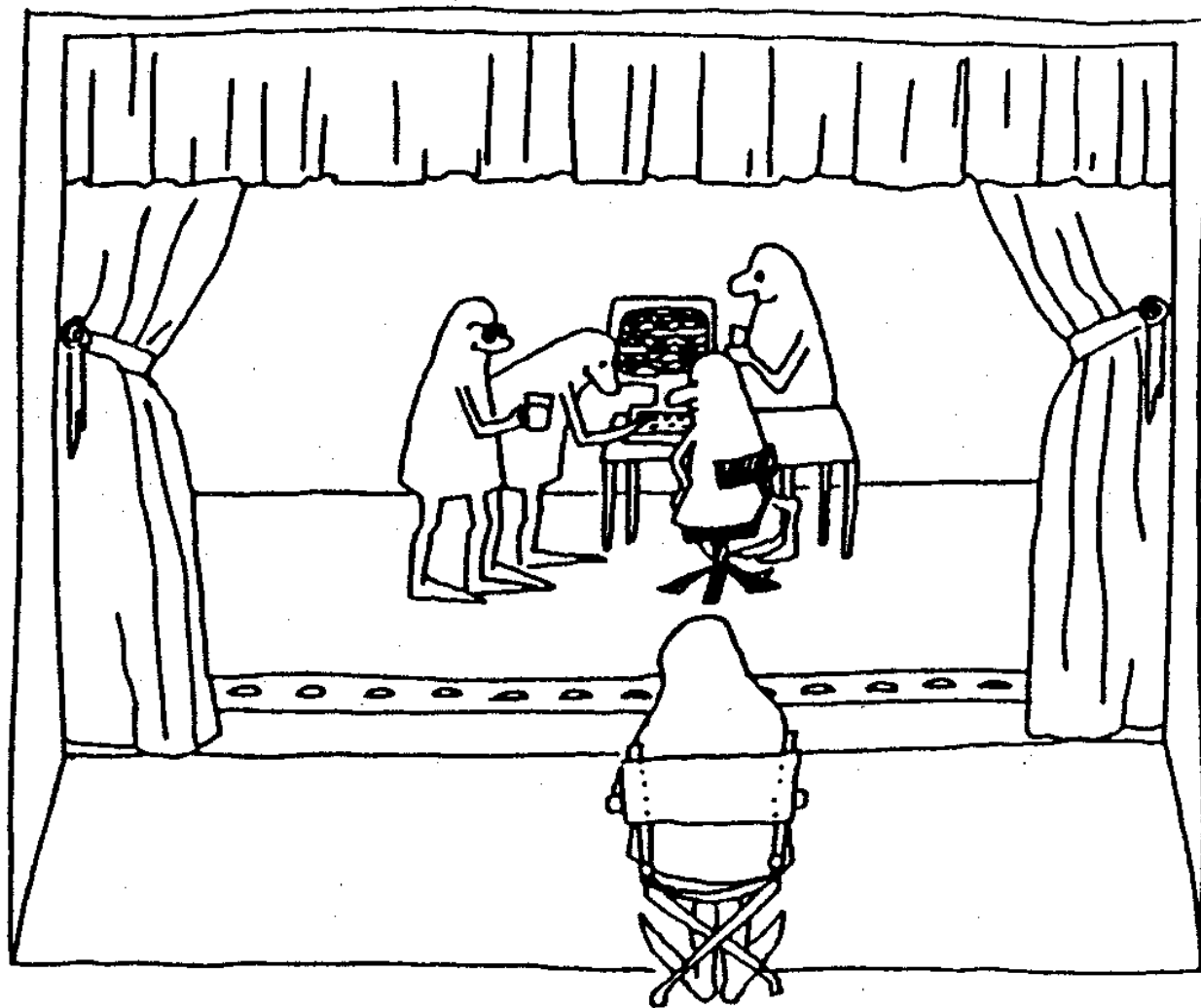
**People use
computers**

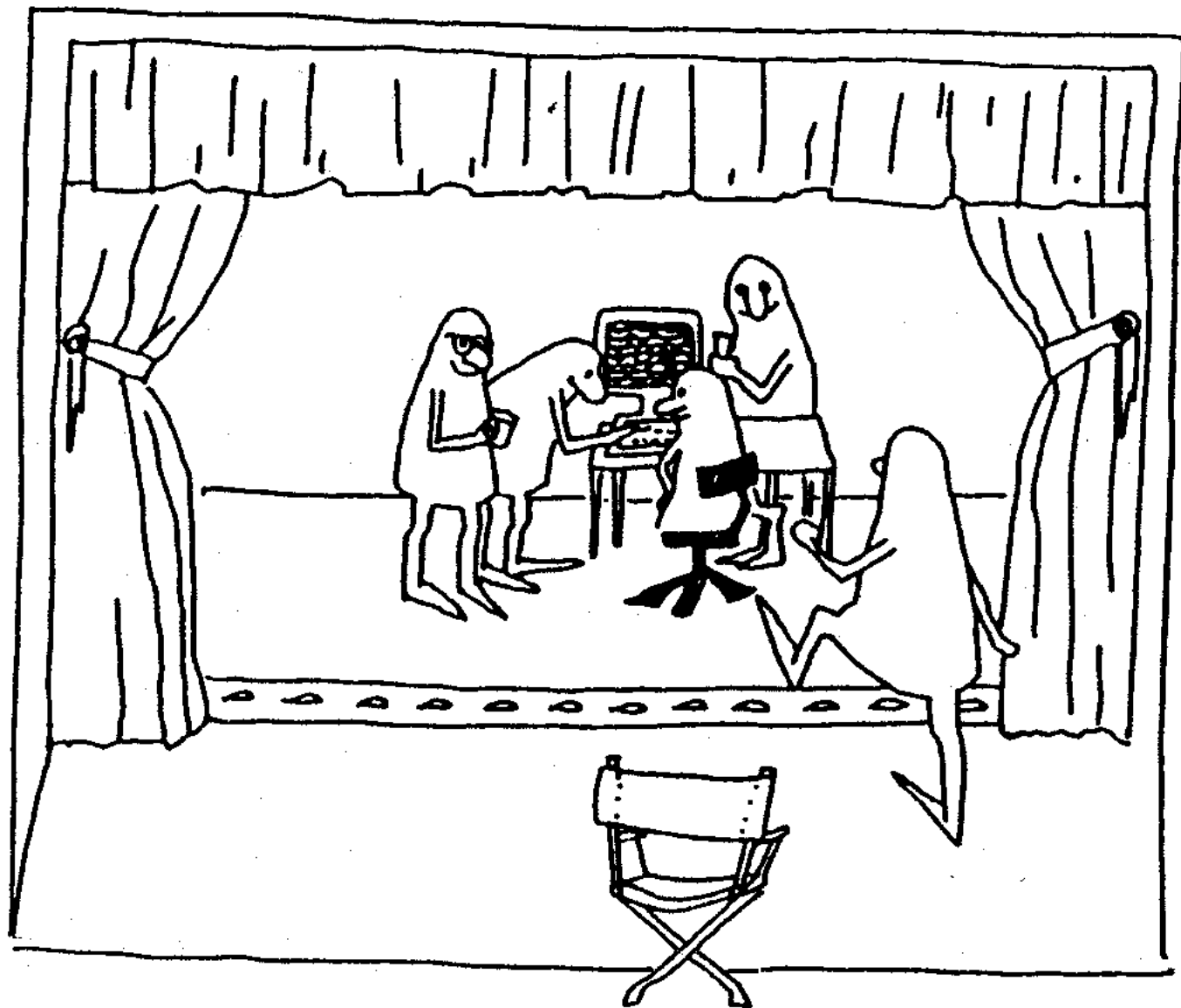
**I test computers when people
use them**

Participatory design

people destined to use the system play a critical role in *designing it*.

Schuler & Namioka, 1993, p. xi





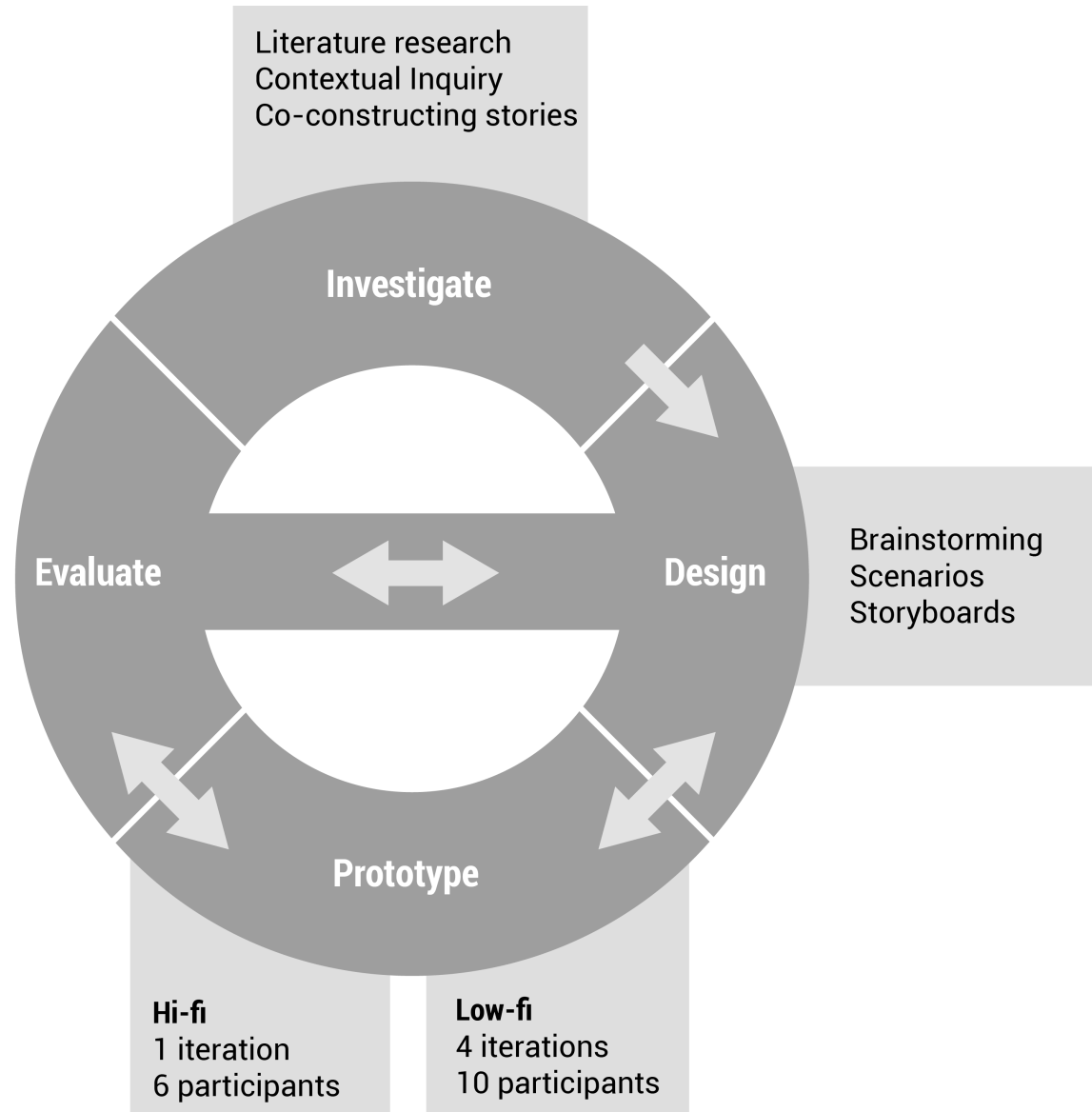
XR are perceptual systems

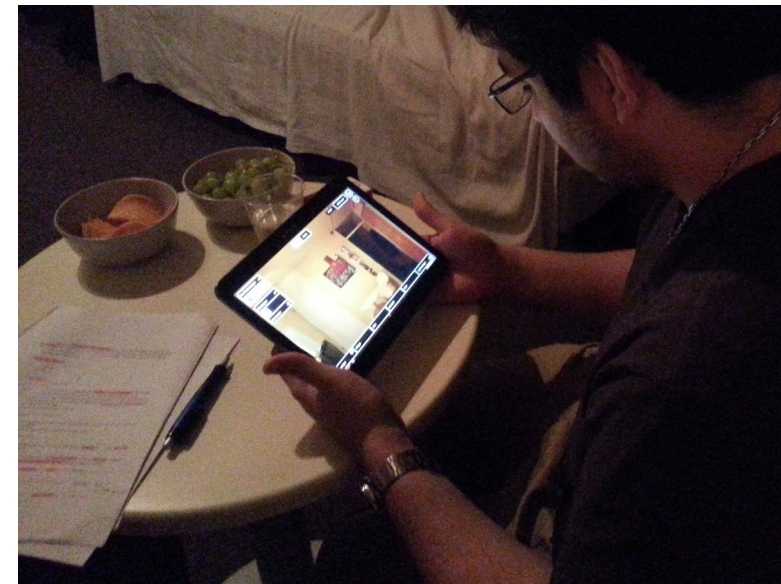
They cannot exist without a human in the loop to perceive what is happening

Test fast

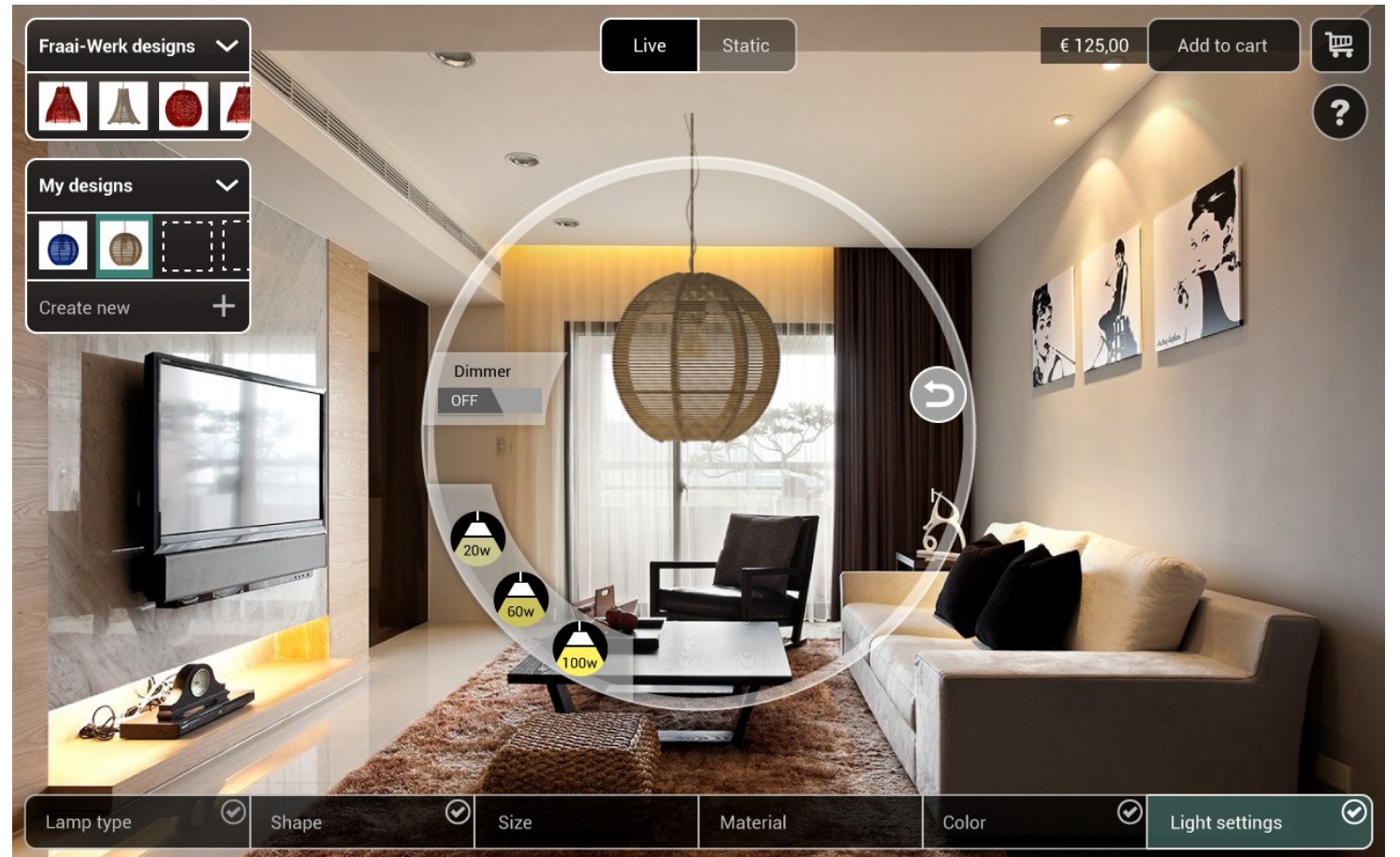
Is it possible in XR?



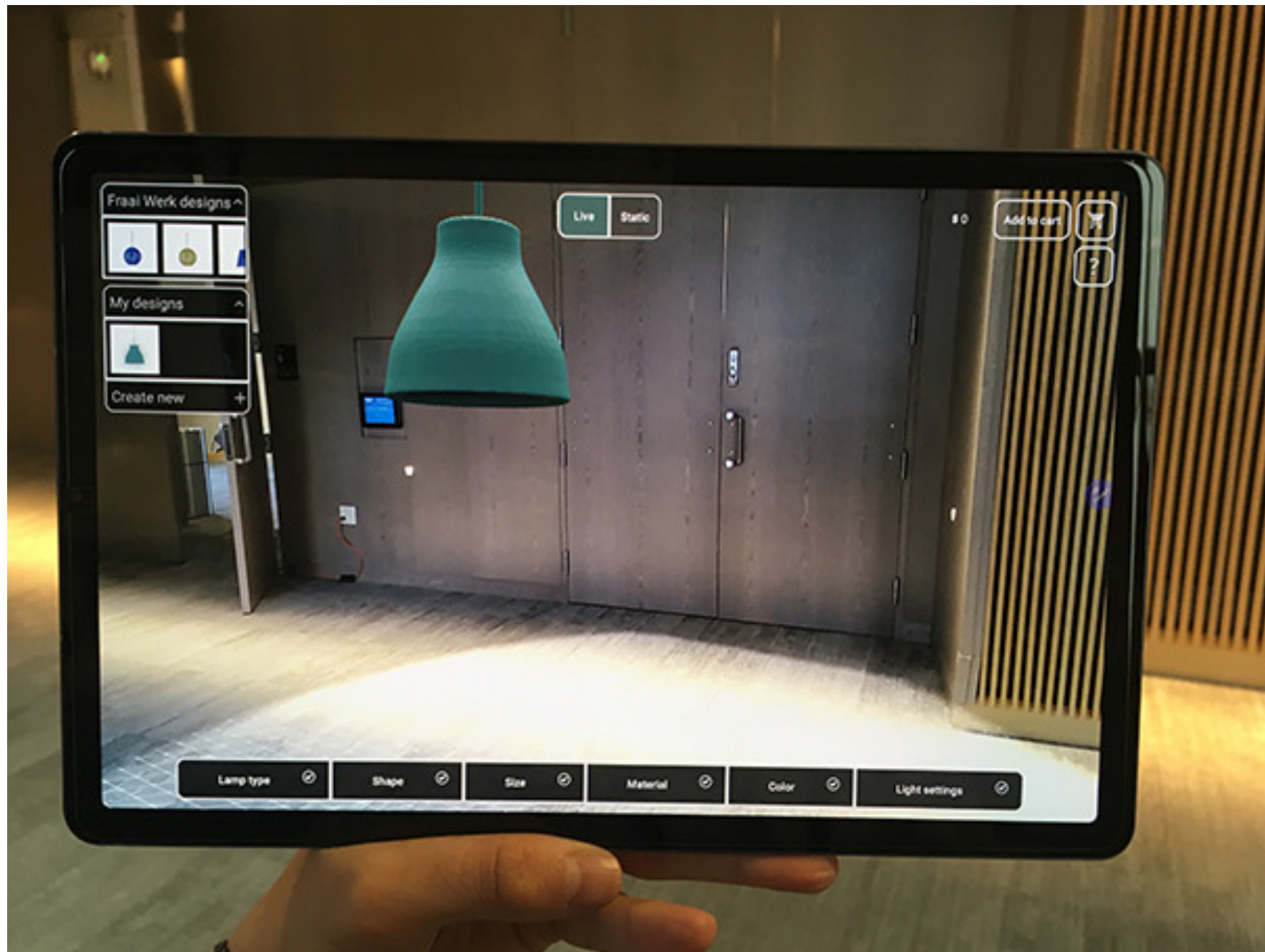




Perusquía-Hernández, M., Kriening, H., Palumbo, C., Wajda, B. User-Centered Design of a Lamp Customization Tool. Proceedings of the 5th Augmented Human International Conference, Article 36, 2 pages. 2014. DOI:10.1145/2582051.2582087



To-keep-thinking...

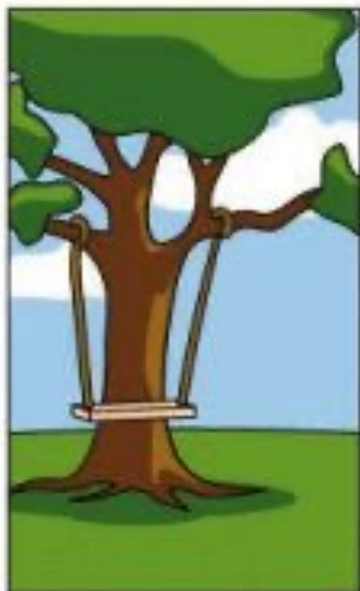


**Prototypes are not only
good for testing**

**They are also a
communication tool**



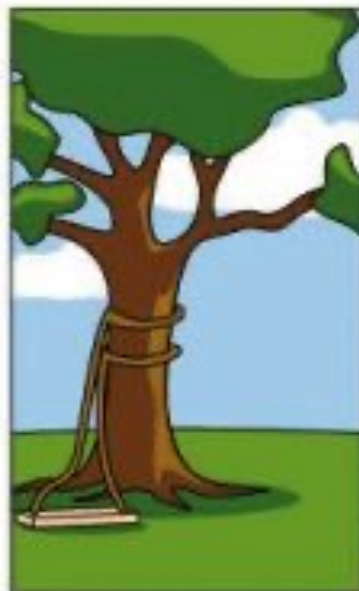
How the customer explained it



How the Project Leader understood it



How the Analyst designed it



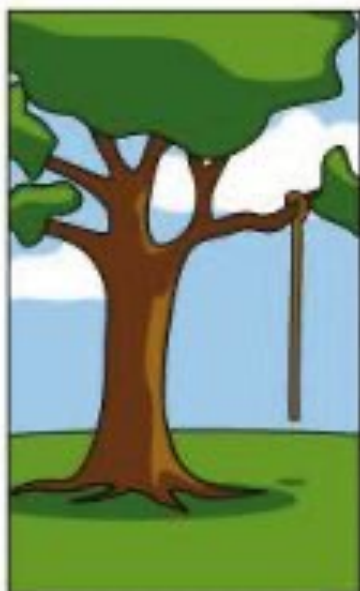
How the Programmer wrote it



How the Business Consultant described it



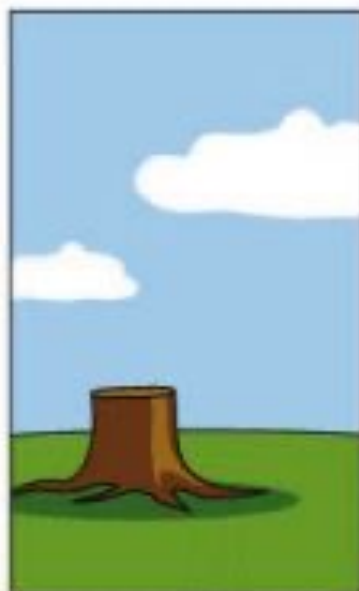
How the project was documented



What operations installed



How the customer was billed



How it was supported



What the customer really needed

Time



How the customer explained it



How the Project Leader understood it



How the Analyst designed it



How the Programmer wrote it



How the Business Consultant described it



How the project was documented



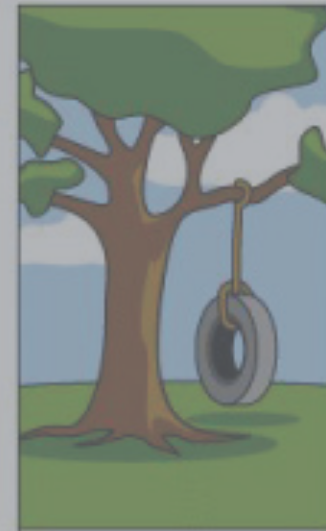
What operations installed



How the customer was billed



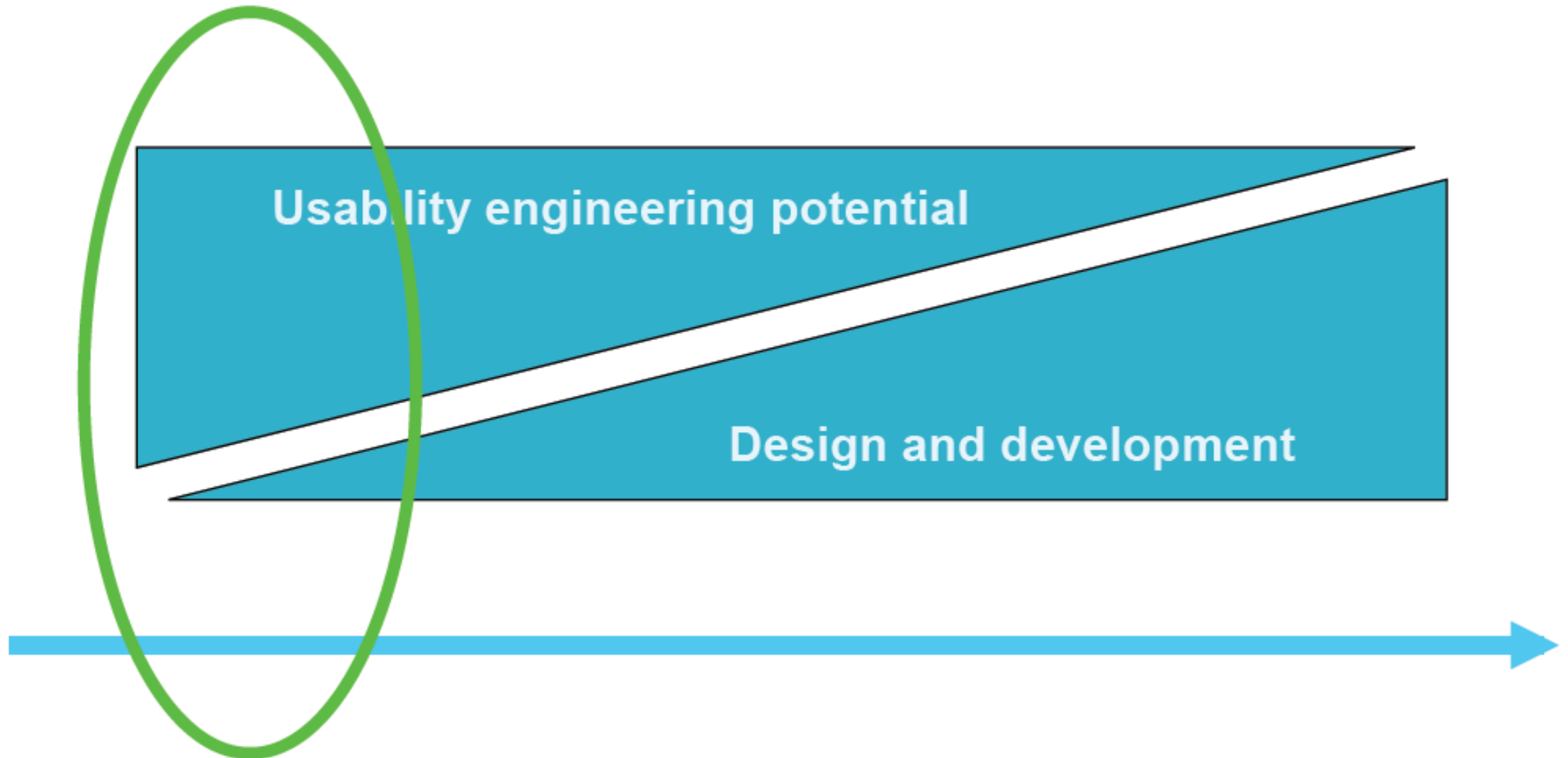
How it was supported



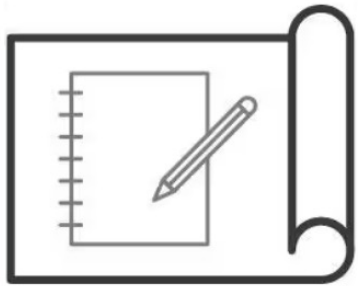
What the customer really needed

Money

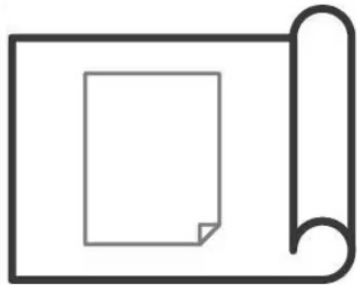
Usability effort can be more efficient at the start of the project



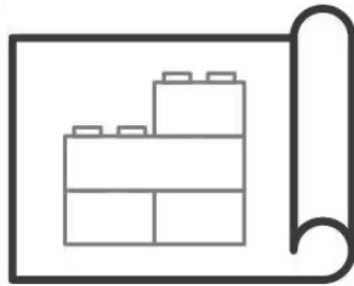
5 Common Low-Fidelity Prototypes



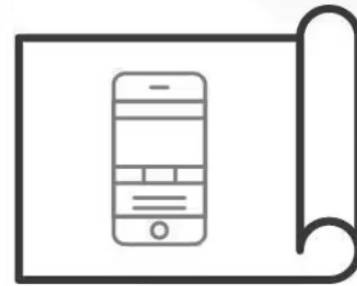
Sketches



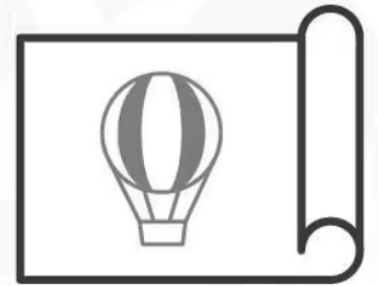
Paper



Lego



Digital



Wizard of Oz



Prototype as soon as you can

Storyboards, concept drawings, anything in paper is useful

Diver's Point of View (Through XR-enabled Mask)

Guide Fish Avatar



Nearby buddies
(with distance)

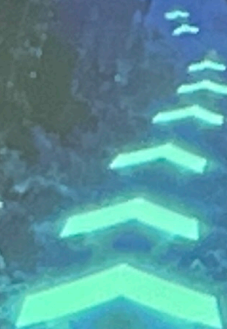
Diver B
12 m



Diver C
16 m



Guided route
(breadcrumbs)



⚠ Strong Current
- Keep Right

⚠ Sharp Object
Avoid

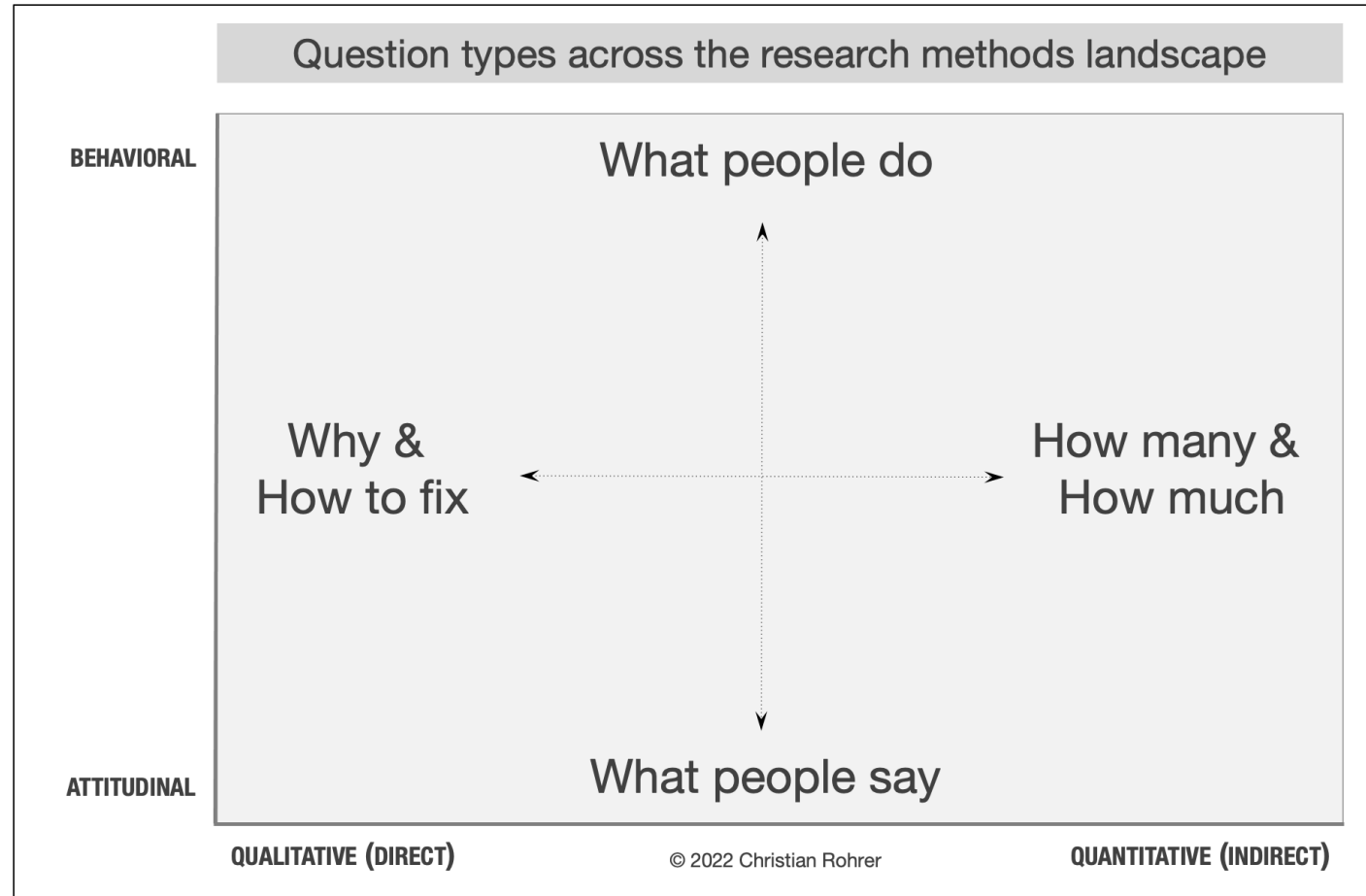
DEPTH
18 m
MAX
30 m

Max Safe Depth: 30 m

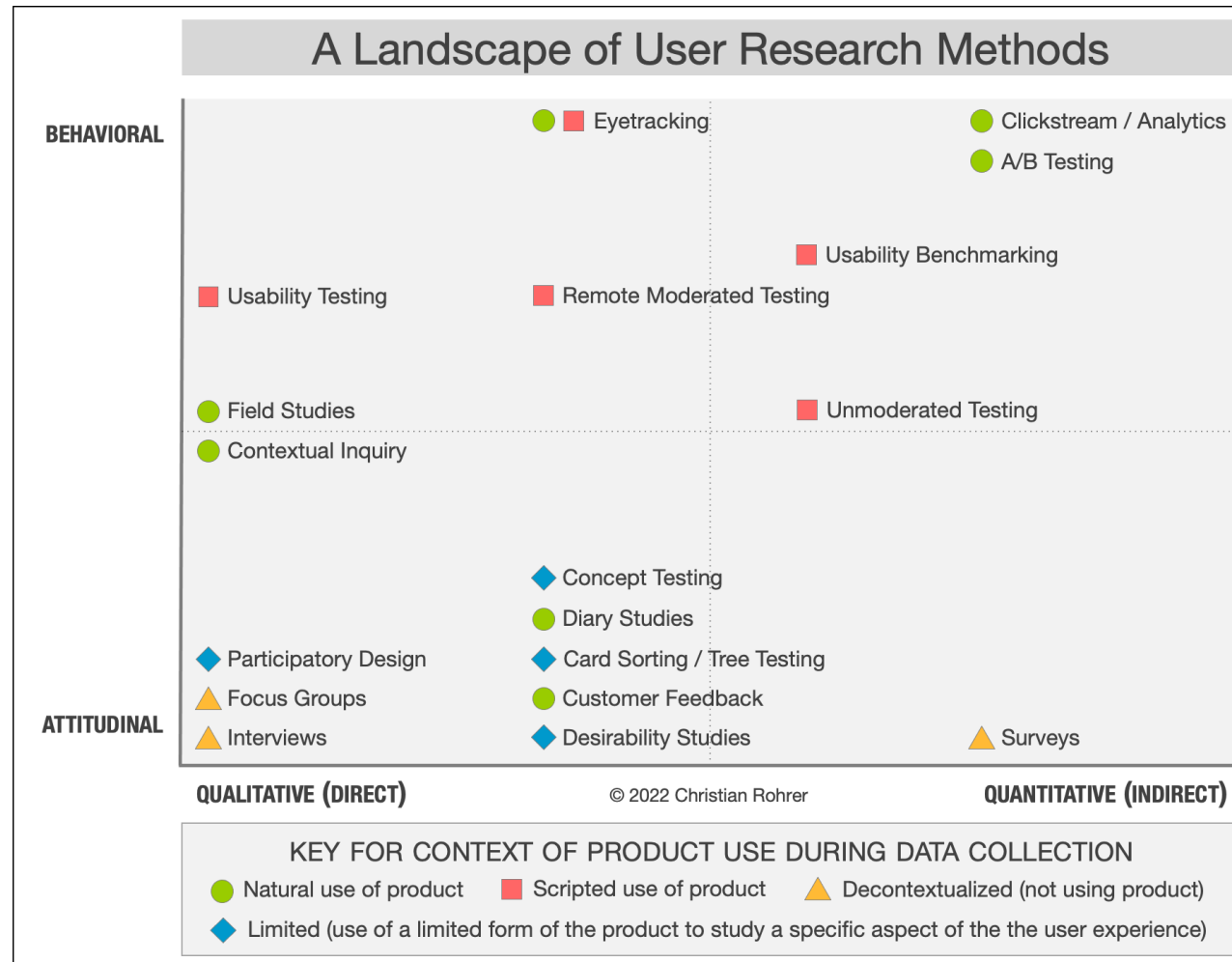


Approaching limit
(5 m remaining)

User study methods



User study methods



Quick tests

- **Heuristic evaluation**

An expert who knows basic design heuristics evaluates a design prototype

<https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/>

- **User evaluation**

Users do a task in a prototype

- Observation
- Thinking aloud
- Simple performance metrics (task finish)

Heuristic evaluation

<https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/>



← **ISSUE:**
**TEXT OVERLAPS
WITH PRODUCT
IMAGES**

NN/g NNGROUP.COM

Nielsen Norman Group

Heuristic Evaluation Workbook

EVALUATOR:

DATE:

PRODUCT:

TASK:

1

Visibility of System Status

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.

- Does the design clearly communicate its state?
- Is feedback presented quickly after user actions?

Issues

Recommendations

2

Match Between System and the Real World

The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order.

- Will user be familiar with the terminology used in the design?
- Do the design's controls follow real-world conventions?

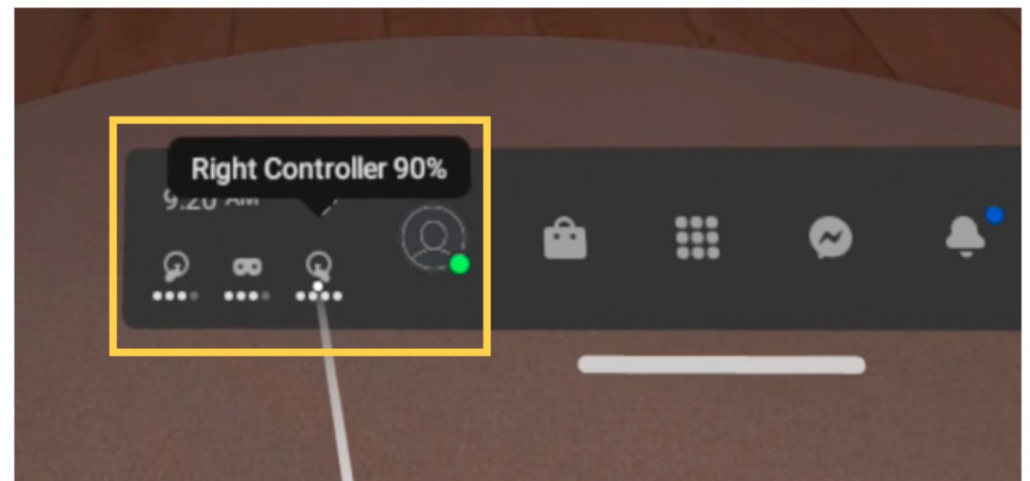
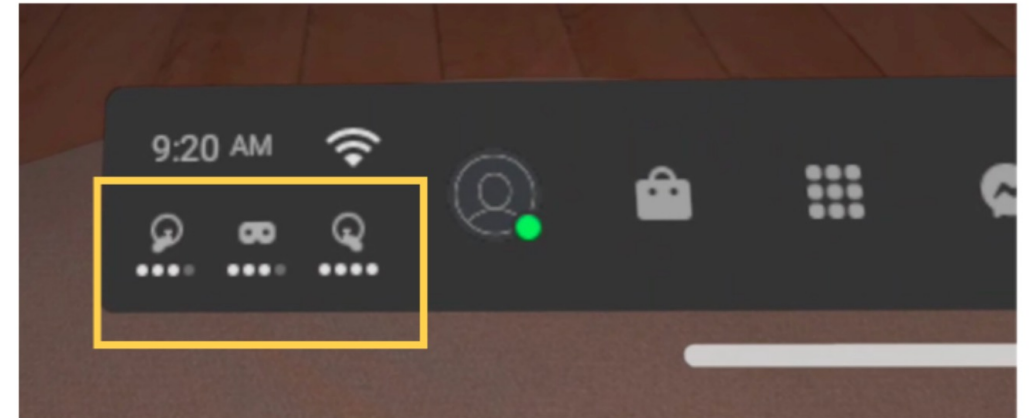
Issues

Recommendations

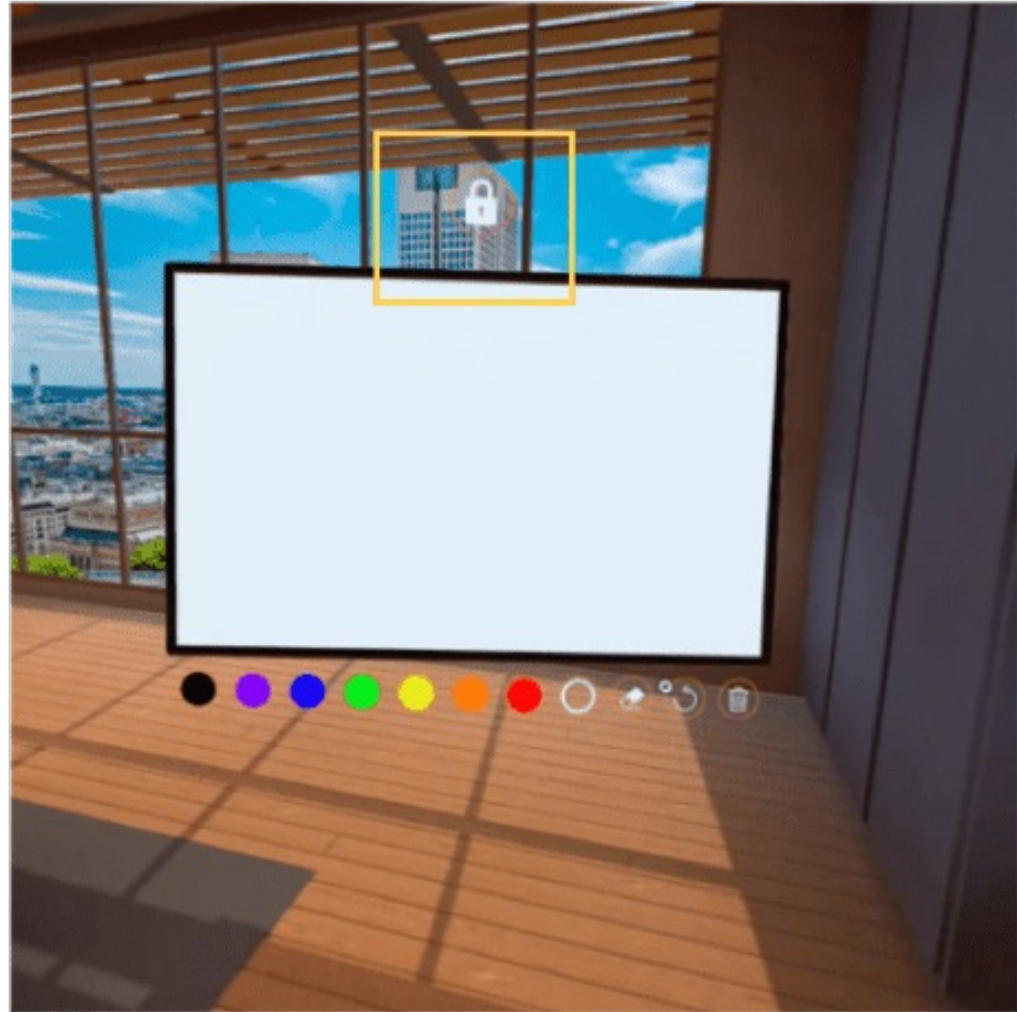
Heuristic evaluation for VR

<https://www.nngroup.com/articles/usability-heuristics-virtual-reality/>

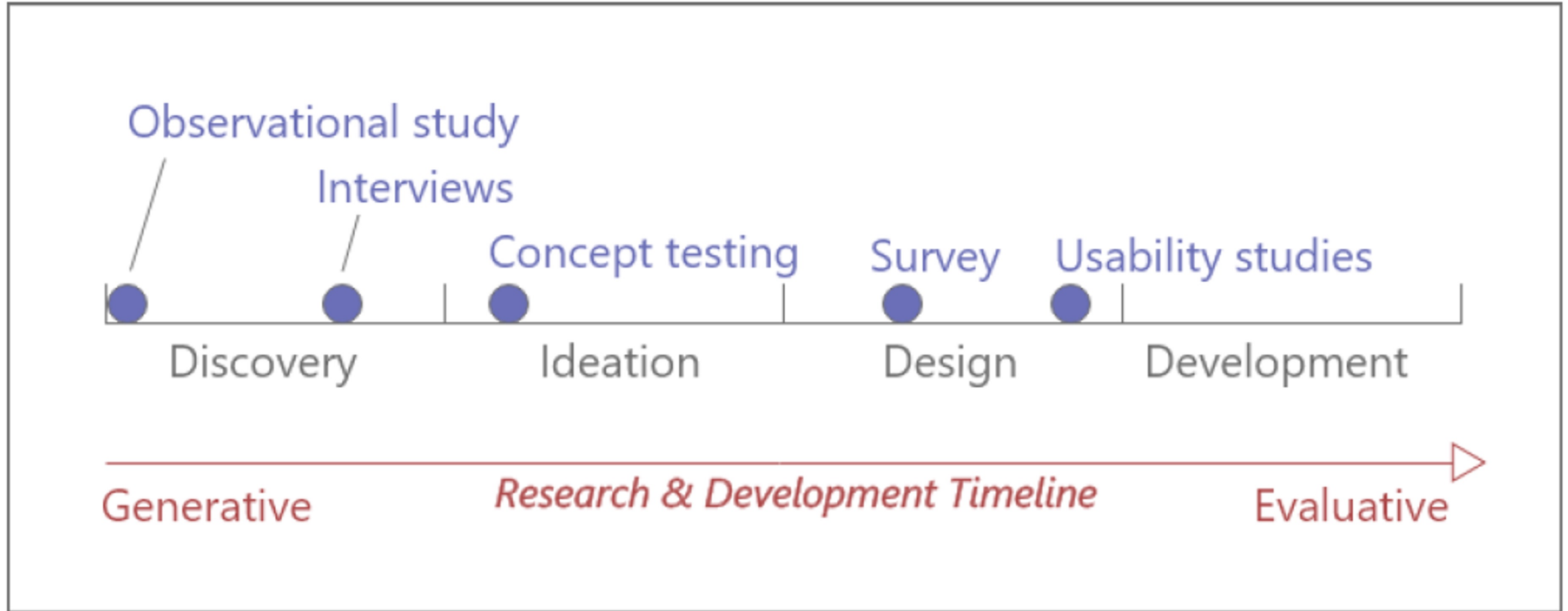
1. Visibility of System Status
2. Match Between System and the Real World
3. User Control and Freedom
4. Consistency and Standards
5. Error Prevention
6. Recognition Rather than Recall
7. Flexibility and Efficiency of Use
8. Aesthetic and Minimalist design
9. Help Users Recognize, Diagnose, and Recover from Errors
10. Help and Documentation



Match with the real world



Generative to evaluative methods



Bylinskii, Z., Herman, L., Hertzmann, A., Hutka, S., & Zhang, Y. (2022). Towards Better User Studies in Computer Graphics and Vision. *arXiv preprint arXiv:2206.11461*.

Questions?

Rigorous evaluation for publication

Experiments are necessary

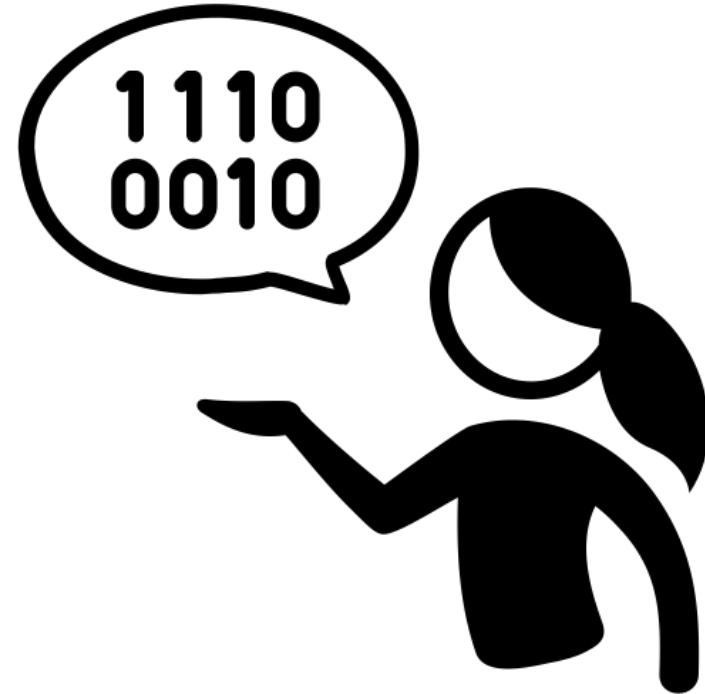
Contents

もくじ

- Why do CS need experiments with people?
- **Reproducibility crisis and Open Science**
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Does it work?

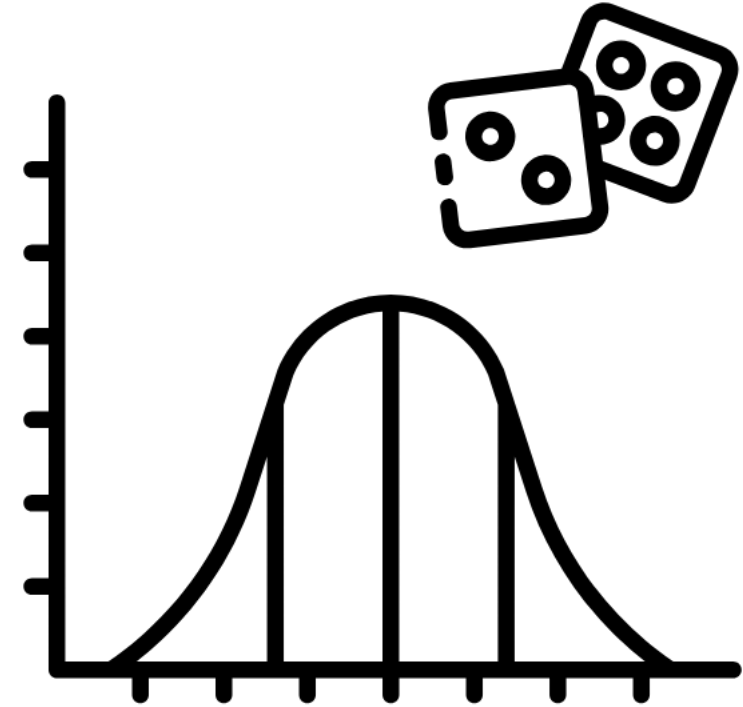
Yes or No



**If only it were
this easy**

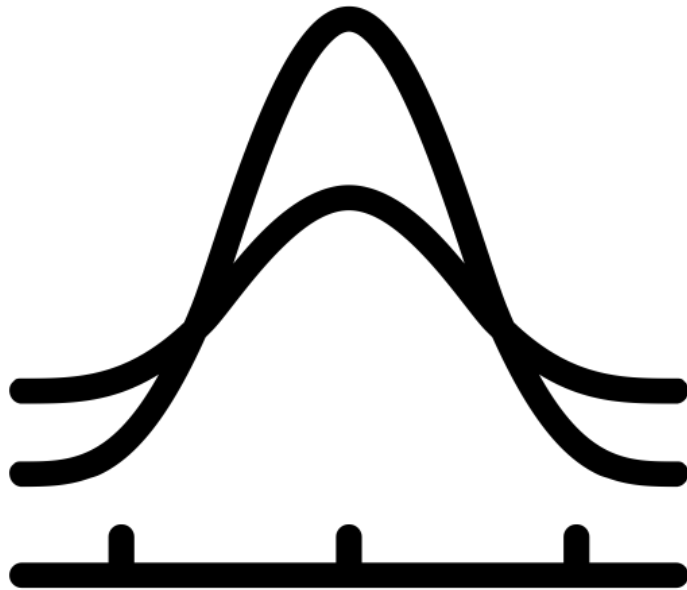
Are people going to use it without errors?

Probability distribution

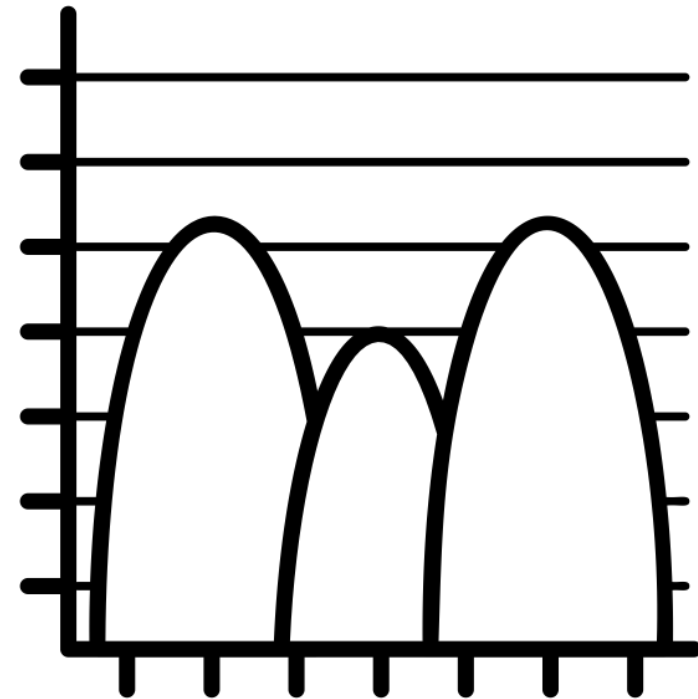


Created by Andi Nur Abdillah
from Noun Project

Is a joystick better than a keyboard?



Created by icon trip
from Noun Project



Created by Vectors Point
from Noun Project



IS THERE A REPRODUCIBILITY CRISIS IN SCIENCE?

Open Science Framework

Manage and share your research with OSF – an easy, integrated platform.

OSF is a free and open source project management tool that supports researchers throughout their entire project lifecycle.

As a collaboration tool, OSF helps research teams work on projects privately or make the entire project publicly accessible for broad dissemination. As a workflow system, OSF enables connections to the many products researchers already use, streamlining their process and increasing efficiency.

Structured projects

Manage files, data, code, and protocols in one centralized location and easily build custom organization for your project — no more trawling emails to find files or scrambling to recover lost data.

Controlled access

Manage which parts of a project are public or private, making it easy to collaborate and share with the community or just your team.

Enhanced workflow

Connect your favorite third-party services like Dropbox or Google Drive, automate version control, preregister your research, share preprints.

Dependable repository

OSF's preservation fund is sufficient for 50+ years of read access hosting at present costs.

Landing page:

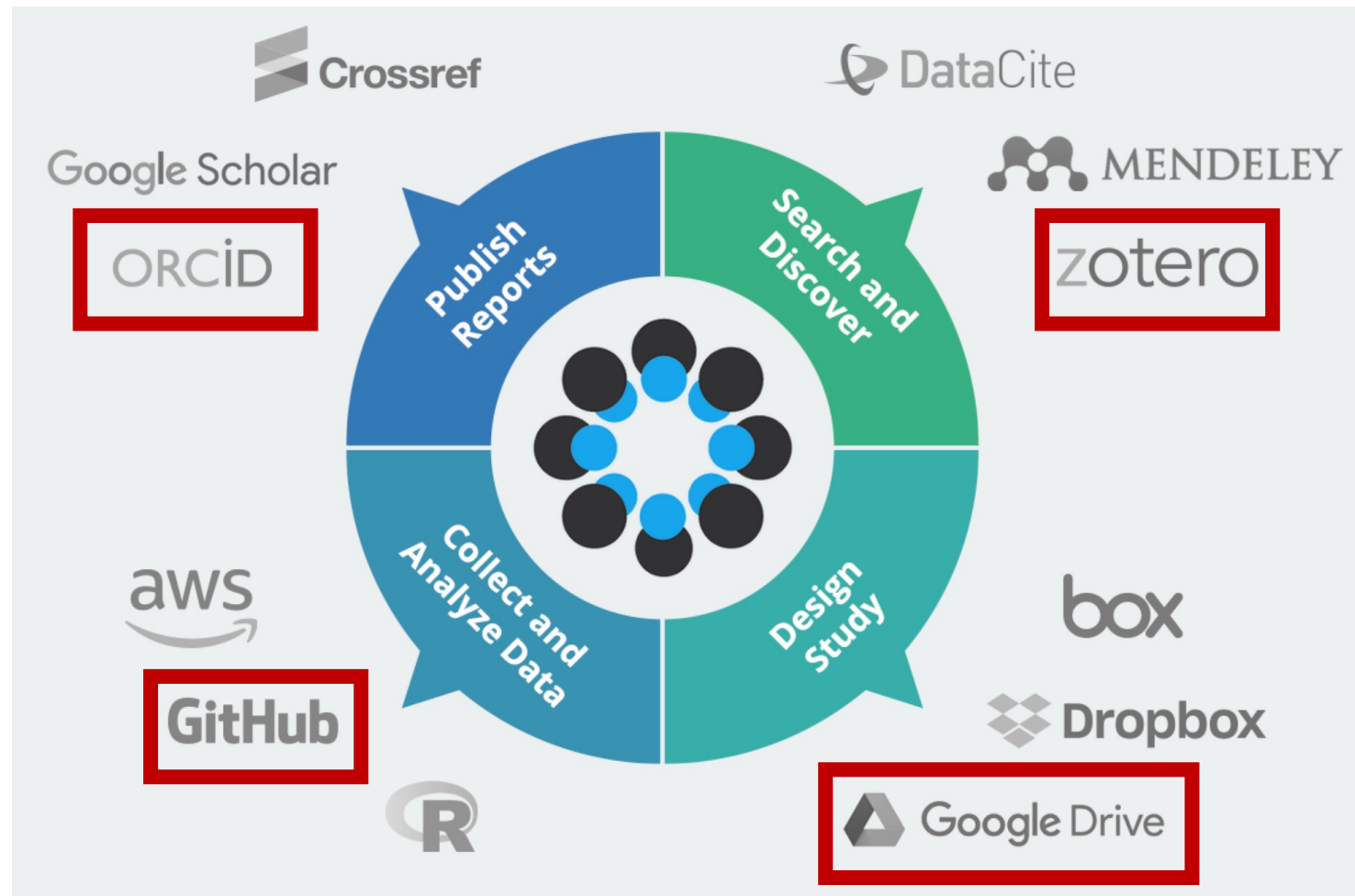
<https://www.cos.io/>

Lectures:

<https://www.cos.io/events>

Get started now

Open Science Framework



Tools to manage the research cycle



ScienceVerse

15 followers <http://www.scienceverse.org>

README.md

scienceverse

The increasingly digital workflow in science has made it possible to share almost all aspects of the research cycle, from pre-registered analysis plans and study materials to the data and analysis code that produce the reported results. Although the growing availability of research output is a positive development, most of this digital information is in a format that makes it difficult to find, access, and reuse. A major barrier is the lack of a framework to concisely describe every component of research in a machine-readable format: A grammar of science.

Lakens, D., & DeBruine, L. M. (2021). Improving Transparency, Falsifiability, and Rigor by Making Hypothesis Tests Machine-Readable. *Advances in Methods and Practices in Psychological Science*. <https://doi.org/10.1177/2515245920970949>

People



Top languages

R HTML TeX SCSS

[Report abuse](#)

<https://github.com/scienceverse>

Collaboration and sharing

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RRR: A Multi-Lab Replication Of The Induced Compliance Paradigm Of Cognitive Dissonance

Public Project

0

Wiki

This is a large scale registered replication report (RRR) of a major cognitive dissonance paradigm: induced compliance with a counterattitudinal essay.

The general goal of this project is to replicate a seminal study from the cognitive dissonance literature. We chose to replicate Study 1 of Croyle and Cooper (1983), which examined the effect of attitude change in the induced compliance paradigm.

To receive more information about the project, please contact David ([david.vaidis@univ-tlse2.fr][1]) or Willem ([w.sleegers@me.com][2]).

Status

...

[Read more](#)

Files Preview

RRR: A Multi-Lab Replication of the Induced Compliance Paradigm of Cognitive Dissona... ▾

OSF Storage

Data	0 B
Documents	0 B
Lab materials	0 B
Manuscript	0 B
Scripts	0 B

Metadata

Contributors

David Vaidis, Willem Sleegers, Daniel Priolo, Alexandre Bran, Florian van Leeuwen, Kenneth DeMarree, Marie-Pierre Fayant, Bjørn Sætrevik, Bruno verschuere, Andrew R. Smith, Kathleen Schmidt, Amélie Gourdon-Kanhukamwe, Rabia Ibrahim Kodapanakkal, John Protzko, Gregory John Depow, Fatih Bayrak, Mioara Cristea, Kevin Vezirian, Armand Chatard, Lisa Moussaoui

[See more](#)

Resource Type

No resource type

Description

A Large Scale Replication Project of Cognitive Dissonance Theory. Part I - Using an Induced Compliance Paradigm and a Counterattitudina...

[Read more](#)

Date Created	Date Updated
Dec 11, 2018, 5:20 PM	Jan 9, 2025, 1:34 AM

Subjects

None

Tags

attitude change cognitive dissonance

Cognitive dissonance theory

Counterattitudinal Essay

Induced Compliance Paradigm

Components

Pre-registration

The **process of documenting** a study's hypotheses, methodology, and analysis plan **in a public, time-stamped registry before data collection begins.**

It is highly valuable because it **prevents bias** (e.g., selectively reporting only positive outcomes), **distinguishes planned research from accidental discoveries**, and **increases the overall credibility and transparency** of scientific findings.

If you preregister, you are not tempted to change half-way to find what is convenient to find

<https://www.cos.io/blog/prereg-for-research-communities>

The Perspective Effect Of Awe In VR Environments: A Comparison Of First-Person And Third-Person Perspectives

Otsubo et al. 2024. First-Person Perspective Induces Stronger Feelings of Awe and Presence Compared to Third-Person Perspective in Virtual Reality. In Proceedings of the 26th International Conference on Multimodal Interaction (ICMI '24). <https://doi.org/10.1145/3678957.3685753>



This is an update to the original registration. This update was made on Nov 6, 2024.

Reason for update:



Data



Analytic Code



Materials



Papers



Supplements

Update in progress



Updates



Preregistration Template From AsPredicted.Org

Data collection

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

Hypothesis

H1: There is a difference in the feeling of awe between 1pp and 3pp.

Previous studies suggest that first-person perspective (1pp) would elicit more awe than third-person perspective (3pp), because first-person perspective enhances the sense of presence in VR (Iriye & St. Jacques, 2021). In contrast, we expect that a 3pp would elicit more awe than 1pp, because a 3pp improves spatial cognition and allows one to compare one's own smallness with the vastness of the landscape (Goris et al., 2017).

H2: Feeling awe is related to heart rate reduction.

We base this hypothesis on a study that used two videos, one of awe and one of neutral, to examine how the subjective reports of awe correlate with heart rate (Bernstein, 2022). A previous study found that the mean heart rate was lower during the awe video than during the neutral video, and also that the higher the self-reported level of awe, the lower the mean heart rate level during the awe-inducing video.

H3: Feeling awe is related to speech rate increase.

Metadata

Edit

Contributors

[Hiromu Otsubo](#), [Marvin Lehnort](#),
[Melissa Steininger](#),
[Alexander Marquardt](#), [Felix Dollack](#),
[Yutaro Hirao](#),
[Monica Perusquía-Hernández](#),
[Hideaki Uchiyama](#), [Ernst Kruijff](#),
[Bernhard Riecke](#)

See more

Resource Type

No resource type

Description

Nature scenes like landscapes have been used in several studies to evoke awe.

79

Help

Questions?

Contents

もくじ

- Why do CS need experiments with people?
- Reproducibility crisis and Open Science
- **Experiment protocols and Experiment design**
- Power analysis
- Data analysis using R
- Ethical approval



Experiment protocol template

実験計画のテンプレート



English:

https://docs.google.com/document/d/118Crarfw62Z1odOkCYw2bjbrqtM4fQ_NtFUstNjxic/edit?usp=sharing



Japanese:

<https://docs.google.com/document/d/1RrFIGOOfHreyHFAYICkkuOFIdev8qmTUfftALoyevYg/edit?usp=sharing>



Sugar



Tastiness

Experiment protocol

Title of the experiment (example: Effect of sugar on brownie tastiness)

Date first draft: 2022.04.12 Written by: PERUSQUIA-HERNANDEZ Monica with some explanations from

<https://osf.io/zab38/wiki/home/>

Date revision: _____ By: _____ Changes made: _____

Date revision: _____ By: _____ Changes made: _____

Background of the experiment 実験の背景

Please give a brief description of your study, including some background, the purpose of the study, or broad research questions.

Provide a quick summary of the literature background with references, and why your study is covering a gap in the literature.

Example: Though there is strong evidence to suggest that sugar affects taste preferences, the effect has never been demonstrated in brownies. Therefore, we will measure taste preference for four different levels of sugar concentration in a standard brownie recipe to determine if the effect exists in this pastry.

Societal contribution 社会への貢献

What do you expect to learn from this experiment? How do the findings contribute to science and society?

Example: By finding a balance between a brownie's sugar and preference, we can avoid unnecessarily high sugar levels, thereby reducing the incidence of sugar-related diseases.



Brownie
Sugar



Tastiness

Experiment protocol

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Date first draft: 2022.04.12 Written by: PERUSQUIA-HERNANDEZ Monica with some explanations from
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Societal contribution 社会への貢献

What do you expect to learn from this experiment? How do the findings contribute to science and society?

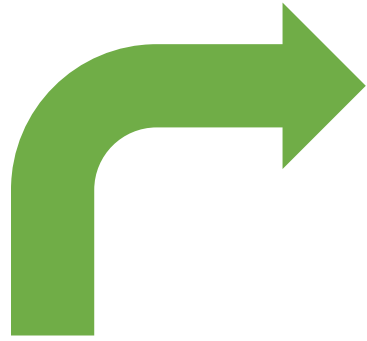
Example: By finding a balance between a brownie's sugar and preference, we can avoid unnecessarily high sugar levels, thereby reducing the incidence of sugar-related diseases.

Hypothesis 仮説

A proposed explanation or educated guess for a phenomenon, often acting as a starting point for scientific investigation

Core Characteristics:

- **Testable:** You must be able to measure and observe it through experiments or data collection
- **Falsifiable:** There must be a possibility to prove the hypothesis wrong
- **Specific:** It outlines exactly what you expect to happen between variables



**How does sugar in brownies
affects their tastiness?**

Research question

The hypothesis is an expected answer

Hypothesis

What do you expect to discover from this experiment?

List specific, concise, and testable hypotheses. Please state if the hypotheses are directional or non-directional. If directional, state the direction. A predicted effect is also appropriate here. If a specific interaction or moderation is important to your research, you can list that as a separate hypothesis.

Example: If taste affects preference, then mean preference indices will be higher with higher concentrations of sugar.

**How does sugar in brownies
affects their tastiness?**

Hypothesis

If taste affects preference, then mean preference indices will be higher with higher concentrations of sugar.

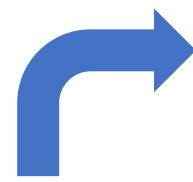
It is an educated guess; please add a rationale from the literature

H1: There is a difference in the feeling of awe between 1pp and 3pp.

Previous studies suggest that first-person perspective (1pp) would elicit more awe than third-person perspective (3pp), because first-person perspective enhances the sense of presence in VR (Iriye & St. Jacques, 2021). In contrast, we expect that a 3pp would elicit more awe than 1pp, because a 3pp improves spatial cognition and allows one to compare one's own smallness with the vastness of the landscape (Gorisse et al., 2017).

Otsubo et al. 2024. First-Person Perspective Induces Stronger Feelings of Awe and Presence Compared to Third-Person Perspective in Virtual Reality. In Proceedings of the 26th International Conference on Multimodal Interaction (ICMI '24). <https://doi.org/10.1145/3678957.3685753>

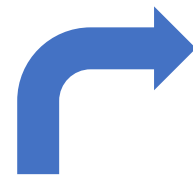
Hypothesis

 **Dependent variable**

If taste affects preference, then **mean preference indices** will be higher with higher **concentrations of sugar**.

 **Independent variable**

Hypothesis

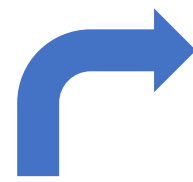
 Dependent variable

If taste affects preference, then **mean preference indices** will be higher with higher **concentrations of sugar**.

 Prediction

 Independent variable

Hypothesis

 Dependent variable (DV)

If taste affects preference, then **mean preference indices** will be higher with higher concentrations of sugar.

 Prediction

 Independent variable (IV)

- Each hypothesis has **only one IV and one DV**
- Each hypothesis has **a statistical test associated** with it (testable)

Variables

変数

Variables

Describe all variables (both manipulated and measured variables) that will later be used in your confirmatory analysis plan. In your analysis plan, you will have the opportunity to describe how each variable will be used. If you are measuring variables for exploratory analyses, you are not required to list them, but it is recommended.

Manipulated variables (independent variables or experimental conditions)

- Describe all variables you plan to manipulate and the levels or treatment arms of each variable. This is not applicable to any observational study.
- **Example:** We manipulated the percentage of sugar by mass added to brownies. The four levels of this categorical variable are: 15%, 20%, 25%, or 40% cane sugar by mass.
- **More information:** For any experimental manipulation, you should give a precise definition of each manipulated variable. This must include a precise description of the levels at which each variable will be set, or a specific definition for each categorical treatment. For example, “loud or quiet,” should instead give either a precise decibel level or a means of recreating each level. ‘Presence/absence’ or ‘positive/negative’ is an acceptable description if the variable is precisely described.

Measured variables (dependent variables – measured subjective states (questionnaires) or sensed events)

- Describe each variable that you will measure. This will include outcome measures, as well as any predictors or covariates that you will measure.
- **Example:** The single outcome variable will be the perceived tastiness of the single brownie each participant will eat. We will measure this by asking participants ‘How much did you enjoy eating the brownie’ (on a scale of 1-7, 1 being ‘not at all’, 7 being ‘a great deal’) and ‘How good did the brownie taste’ (on a scale of 1-7, 1 being ‘very bad’, 7 being ‘very good’).
- **More information:** Observational studies and meta-analyses will include only measured variables. As with the previous questions, the answers here must be precise. For example, ‘intelligence,’ ‘accuracy,’ ‘aggression,’ and ‘color’ are too vague. Acceptable alternatives could be ‘IQ as measured by Wechsler Adult Intelligence Scale,’ ‘percent correct,’ ‘number of threat displays,’ and ‘percentage of reflectance at 400 nm.’

Independent variables

操作変数（独立変数または実験条件）

If taste affects preference, then mean preference indices will be higher with higher **concentrations of sugar**.



Independent variable (IV)

**Levels = experiment condition
only when there is only one IV**

**Levels (cane
sugar by mass):**

- 1. 15%**
- 2. 20%**
- 3. 25%**
- 4. 40%**

Dependent Variables or Measurements 測定変数

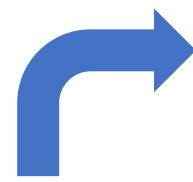
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**Two questionnaires to measure
preference (one variable)**

OK?

Hypothesis

 Dependent variable (DV)

If taste affects preference, then **mean preference indices** will be higher with higher concentrations of sugar.

 Prediction

 Independent variable (IV)

- Each hypothesis has **only one IV and one DV**
- Each hypothesis has **a statistical test associated** with it (testable)

Aggregated variables

指標（または集計変数）

A combination of measurements in a single variable

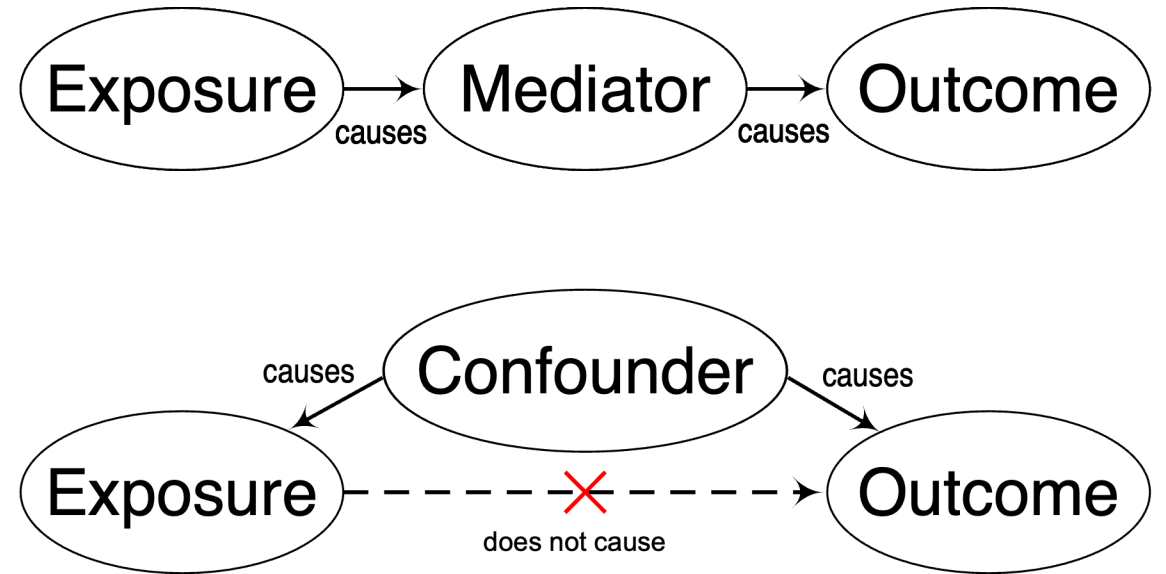
Indices (or aggregated variables)

- If any measurements are going to be combined into an index (or even a mean), what measures will you use and how will they be combined? Include either a formula or a precise description of your method. If you are using a more complicated statistical method to combine measures (e.g., factor analysis), you can note that here, but describe the exact method in the analysis plan section.
- **Example:** We will take the mean of the two questions above to create a single measure of 'brownie enjoyment.'
- **More information:** If you are using multiple pieces of data to construct a single variable, how will this occur? Both the included data and the formula or weights for each measure must be specified. Standard summary statistics, such as "means," do not require a formula, though more complicated indices require either the exact formula or, if it is an established index in the field, the index must be unambiguously defined. For example, "biodiversity index" is too broad, whereas "Shannon's biodiversity index" is appropriate.

Confounding factors

交絡

- Correlation is not causation
- Other variables not in your design can be "confounding" and affect your experiment



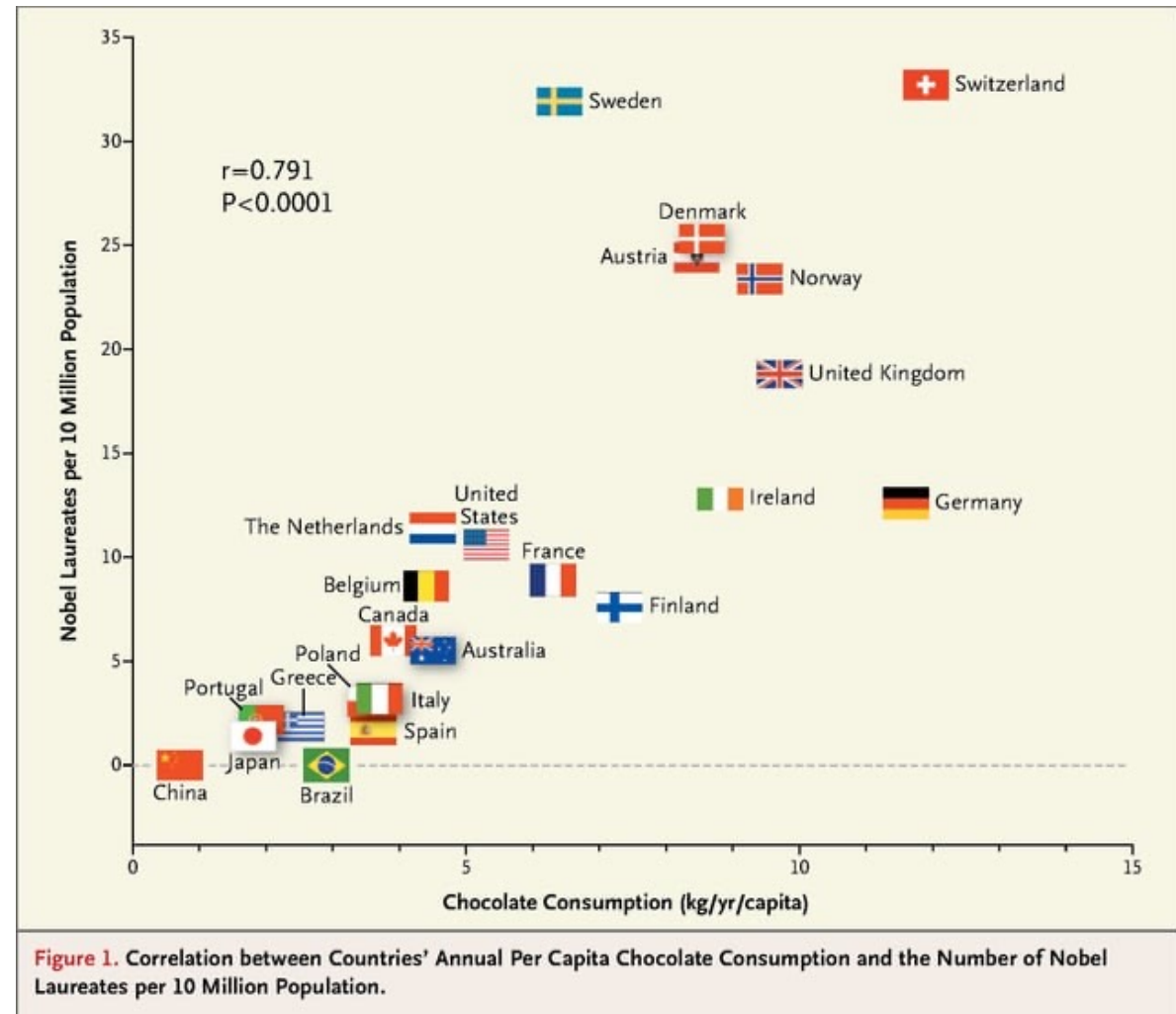
By Cmglee - Own work, CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=90569350>

Confounding factors

交絡

Chocolate consumption is related to number of Nobel Laureates?

Maurage et al. 2013. Does Chocolate Consumption Really Boost Nobel Award Chances? The Peril of Over-Interpreting Correlations in Health Studies. The Journal of Nutrition, Volume 143, Issue 6, Pages 931-933, <https://doi.org/10.3945/jn.113.174813>



Questions?

Experiment design 実験計画

Experimental design

- ☐ Describe your study design. Examples include two-group, factorial, randomized block, and repeated measures. Is it a between (unpaired), within-subject (paired), or mixed design? Describe any counterbalancing required.
- ☐ The most common experiment designs are **between-subjects** (each participant experiences only one condition) and **within-subjects** (each participant experiences all conditions)
- ☐ The type of experimental design determines the type of statistical test that you will perform to validate your hypothesis.
- ☐ If you choose a within-subjects design, the experiment will be longer, but you will need fewer participants. Additionally, you must counterbalance the order in which participants experience each condition.
- ☐ With a between-subjects design, the experiment is shorter, and you prevent learning effects from carrying over between conditions; however, you need more participants to assess your hypothesis.
- ☐ **Example:** We have a between-subjects design with 1 factor (sugar by mass) with 4 levels.

Randomization

- If you are doing a randomized study, how will you randomize, and at what level?
- Example: We will use block randomization, in which each participant is randomly assigned to one of four equally sized predetermined blocks. The random number list used to create these four blocks will be created using the web applications available at <http://random.org>.
- More info: Typical randomization techniques include: simple, block, stratified, and adaptive covariate randomization. If randomization is required for the study, the method should be specified here, not simply the source of random numbers.

Experiment design 実験計画

Between-subjects design

被験者間

- **Different** groups of **participants** receive **different levels** of the IV
- Each participant serves in **only one condition** of the IVs
- **Independent** or **matched samples** are used in each condition

Within-subjects design

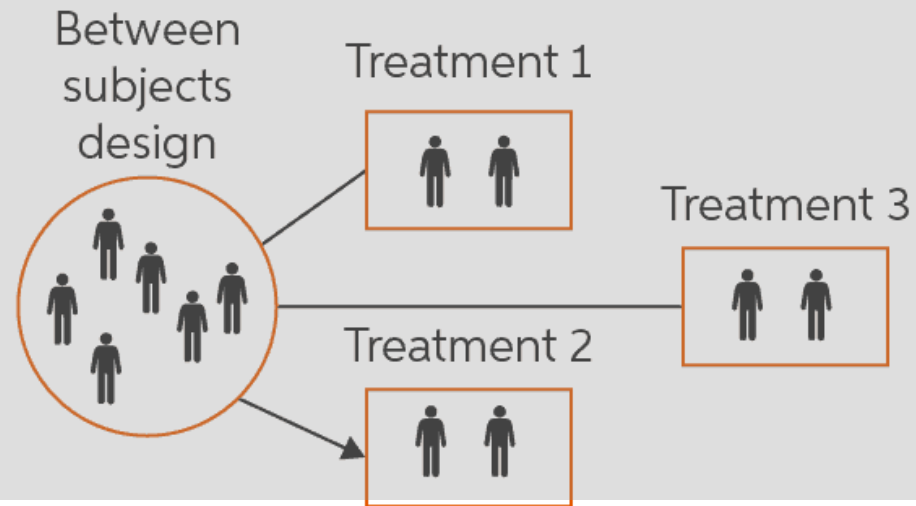
被験者内

- **One group** of participants receives **every level** of the IV
- Each participant serves in **all conditions** of the IVs
- The **same sample** is used in each condition

Experiment design 実験計画

Between-subjects design

被験者間

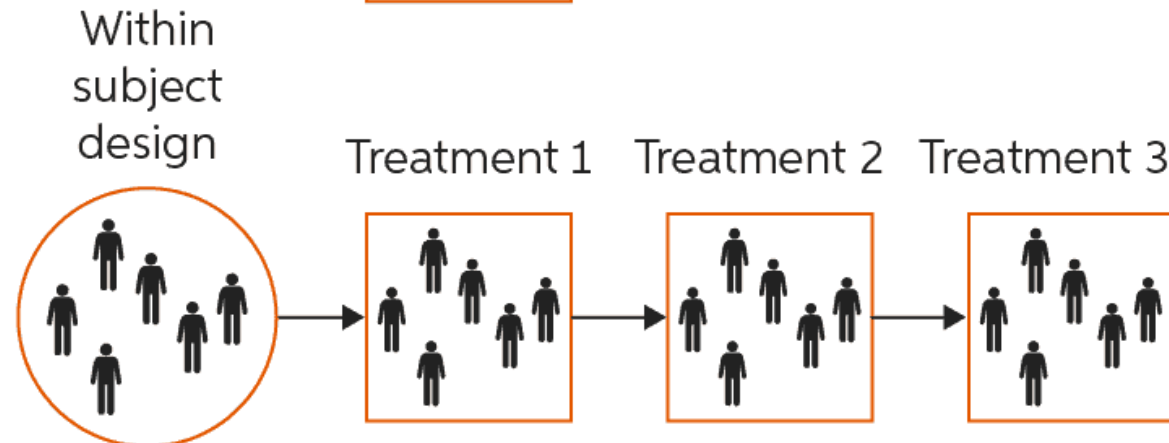


Needs more participants

Conditions are independently measured

Within-subjects design

被験者内



Needs fewer participants

There might be order effects

Experiment design 実験計画

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- More info: Typical randomization techniques include: simple, block, stratified, and adaptive covariate randomization. If randomization is required for the study, the method should be specified here, not simply the source of random numbers.

Counterbalancing

- Randomization might randomly lead to repeated orders
- Counterbalancing is a method to minimize order effects without falling into unwanted imbalances
- The Latin Square method (ラテン方法) is a widely used

columns

	A	B	C	D
rows	B	C	D	A
	C	D	A	B
	D	A	B	C

Each treatment occurs in every column and row

<https://online.stat.psu.edu/stat503/lesson/4/4.3>

https://www.jspt.or.jp/ebpt_glossary/latin.html

Balanced Latin Square Generator

Number of conditions / Square size

6

	1	2	3	4	5	6
1	A	B	F	C	E	D
2	B	C	A	D	F	E
3	C	D	B	E	A	F
4	D	E	C	F	B	A
5	E	F	D	A	C	B
6	F	A	E	B	D	C

Download to .csv

https://damienmasson.com/tools/latin_square/

Stimuli 刺激

Stimuli preparation

- Describe what kind of stimuli you will present to your participants in which condition
- The stimuli can be visual, haptic, auditory, multimodal, or a VR environment.
- If necessary, add links to folders containing the stimuli, or to the GitHub repository containing your implementation



Recipe for the brownies on each condition:

- 1. 15%**
- 2. 20%**
- 3. 25%**
- 4. 40%**

Make sure to give enough detail for it to be reproducible (Open Science Framework)

Taskタスク

Wei, X., Liu, H., Perusquía-Hernández, M., Masai, K., Ioyama, N., Uchiyama, H., Kiyokawa, K. Unobtrusive Visual Acuity Monitoring: Using EOG to Detect Blurred Vision. 45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society. 2023.

Wei, X., Liu, H., Hirao, Y., Perusquía-Hernández, M., Uchiyama, H., Kiyokawa, K. "Electrooculography-Based Detection of Refractive Vision Problems." IEEE Journal of Biomedical and Health Informatics (JBHI). DOI: 10.1109/JBHI.2026.3652204

Task

What exactly will you ask the participants to do? In what order? Does the order change per participant? Please include the counterbalancing table here, if used.

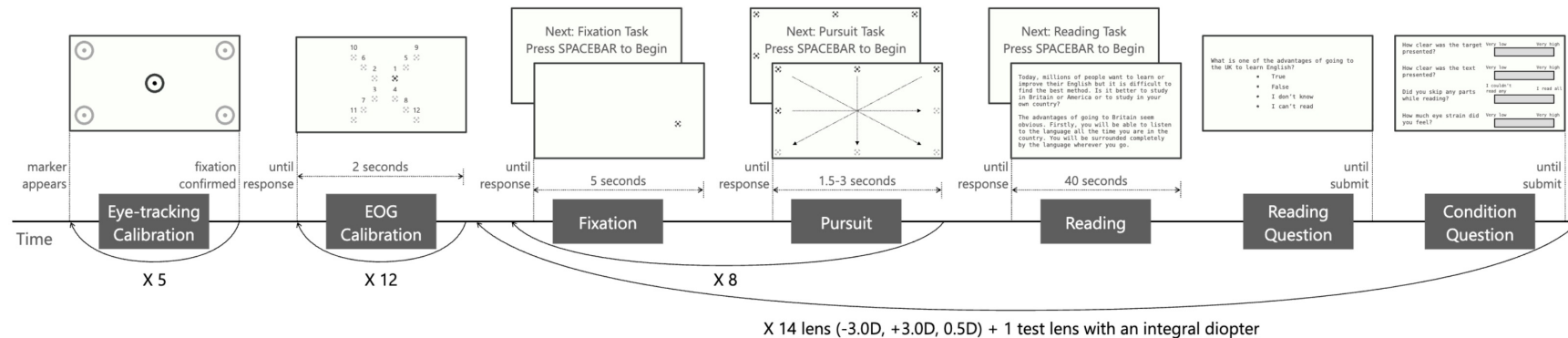


Fig. 1: Flow and task descriptions for the main tasks included in the calibration and experimental conditions of the study. Participants began with the eye-tracking calibration, following a marker on the screen with their eyes. Each marker disappeared once fixation was detected, and a new one appeared at the next position. After five fixations, calibration was complete. This was followed by EOG calibration, where participants fixated on 12 sequentially appearing targets on the screen for two seconds. After calibration, they proceeded with fixation (eight trials, five seconds each), pursuit (eight trials, one and a half to three seconds each), and reading tasks (40 seconds per trial). An attention-check question followed each reading task. Each condition concluded with a questionnaire assessing target clarity, text readability, and eye strain. The experiment included 14 lens conditions: 13 with lens powers from -3.0D to +3.0D in 0.5D steps, plus one additional condition using a repeated integral diopter lens designed to evaluate the model's generalizability in a new context.

Experimental procedure

実験手順

Experimental procedure

Describe the steps that you will follow on the day of the experiment. Include:

- Preparation before the participant arrives (e.g., make sure your XR implementation works, that the logging of the data is working correctly, that you have prepared snacks for the participants, and that you have printed the questionnaires). This preparation is better described as a list with bullet points.
- Preparation while the participant is there (e.g., attach sensors to their body). Provide images to help you remember the exact location of the electrodes.
- What are the steps to follow to explain to the participants?
- What are the exact words that you will tell the participants? – In some psychological experiments, it is better to tell every participant the same instructions, word by word.
- What are the steps to follow after the experiment? Is there data to back up? Where should the data be stored? Is there equipment to clean?

Experimental procedure

実験手順

Wei, X., et al. Unobtrusive Visual Acuity Monitoring: Using EOG to Detect Blurred Vision. EMBC2023.

Wei, X., et al. "Electrooculography-Based Detection of Refractive Vision Problems. JBHI2026

The title of your experiment

ID: _____ Date: _____ Experimenter: _____

Anonymous ID Generation

- The first 2 letters of your first name:
- The first 2 letters of your mother's/female guardian's first name (or XX if not applicable):
- The first 2 letters of your father's/male guardian's first name (or XX if not applicable):
- The number of older siblings you have:
- The day of the month you were born in:
- The last 2 digits of your cell phone number (or 00 if not applicable):

Example

- The first 2 letters of your first name: XI
- The first 2 letters of your mother's/female guardian's first name (or XX if not applicable): QI
- The first 2 letters of your father's/male guardian's first name (or XX if not applicable): RO
- The number of older siblings you have: 0
- The day of the month you were born in: 7
- The last 2 digits of your cell phone number (or 00 if not applicable): 01

The generated anonymous ID is: XIQIRO0701

User Parameter			
Pupillary Distance		Visual Acuity	
Distance from the middle of the eye to the table		Illuminance	
Note			
Calibration		No lens	
Lens #		Lens #	
Lens #		Lens #	

Experimental procedure

実験手順

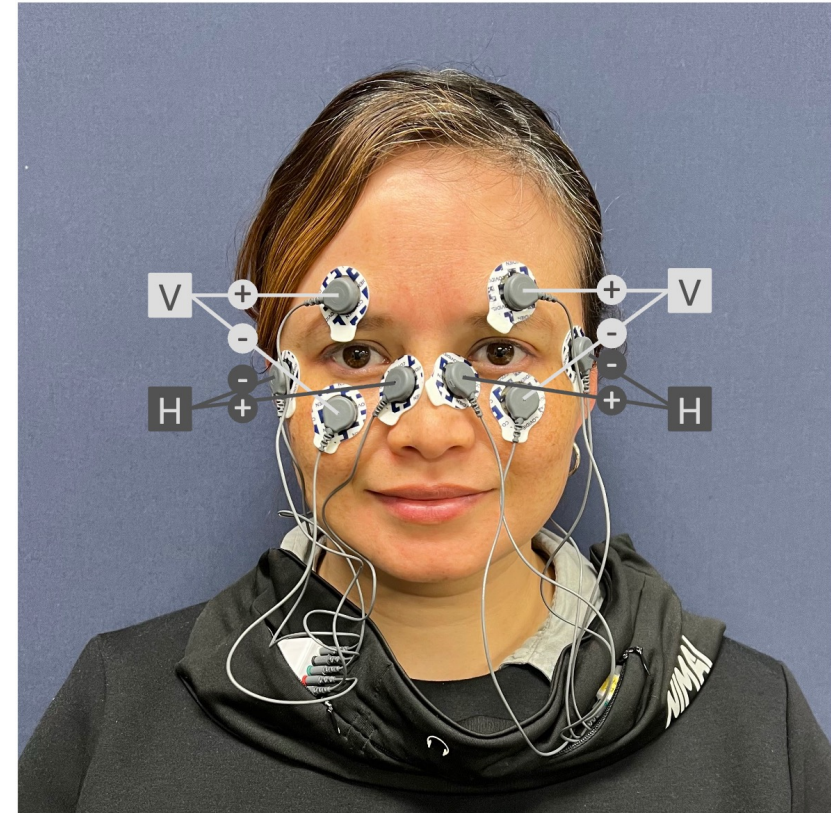
Wei, X., et al. Unobtrusive Visual Acuity Monitoring: Using EOG to Detect Blurred Vision. EMBC2023.

Wei, X., et al. "Electrooculography-Based Detection of Refractive Vision Problems. JBHI2026

ID: _____ Date: _____ Experimenter: _____

Before experiment

- ☐ Laptop is powered
- ☐ check illuminance
- ☐ check trigger
- ☐ check LED light
- ☐ chinrest in place
- ☐ OnePlus 8 internet connection
- ☐ shimmer SD card is empty
- ☐ shimmer battery
- ☐ shimmer configuration
 - ☐ trial name
 - ☐ sampling rate
 - ☐ sensors (ExG: EMG, Proto3: Int A1, slot 5)



Experimental procedure

実験手順

Wei, X., et al. Unobtrusive Visual Acuity Monitoring: Using EOG to Detect Blurred Vision. EMBC2023.

Wei, X., et al. "Electrooculography-Based Detection of Refractive Vision Problems. JBHI2026

During experiment

- ☐ measure pd
- ☐ measure the distance from the middle of the eye to the table
- ☐ explain the experiment
- ☐ sign the documents
- ☐ clean skin
- ☐ attach electrode
 - ☐ mark position according to the eye-tracker
- ☐ check cable connection
- ☐ check data stream by software
- ☐ create a profile on invisible software
- ☐ practice program and adjust the offset
- ☐ change to SDLog firmware
- ☐ check configuration
 - ☐ trial name
 - ☐ sampling rate
 - ☐ sensors (ExG: EMG, Proto3: Int A1, slot 5)
 - ☐ dock / undock to start / stop recording
 - ☐ master -> trigger unit

After experiment

- ☐ All data is ok
- ☐ Write description in shimmer
- ☐ Export data
 - ☐ The SD card is empty after exporting
 - ☐ Backup data in online data storage

Blinding 盲検化

- **Who knows** about the experiment?
- It aims to prevent biases

Blinding

Blinding refers to who is aware of the experimental manipulations in a study. Mark all that apply.

- ☐ No blinding is involved in this study.
- ☐ In studies involving human subjects, participants will not know the treatment group to which they have been assigned.
- ☐ Personnel who interact directly with study subjects (human or non-human) will not be aware of the assigned treatments. (Commonly known as “double blind”)
- ☐ Personnel who analyze the data collected from the study are not aware of the treatment applied to any given group.

Sampling plan サンプルング計画

Sampling plan

Describe **how you plan to collect samples**, as well as the number of samples you plan to collect and your rationale for this decision. Please keep in mind that the data described in this section should be the actual data used for analysis, so if you are using a subset of a larger dataset, please describe the subset that will actually be used in your study.

- **What kind of participants will you invite for your study?**
- **How will you recruit them?**
- **Will you use existing data collected from another study or open databases?**

Questions?

Contents

もくじ

- Why do CS need experiments with people?
- Reproducibility crisis and Open Science
- Experiment protocols and Experiment design
- **Power analysis**
- Data analysis using R
- Ethical approval

Sample size サンプルサイズ

Sampling size

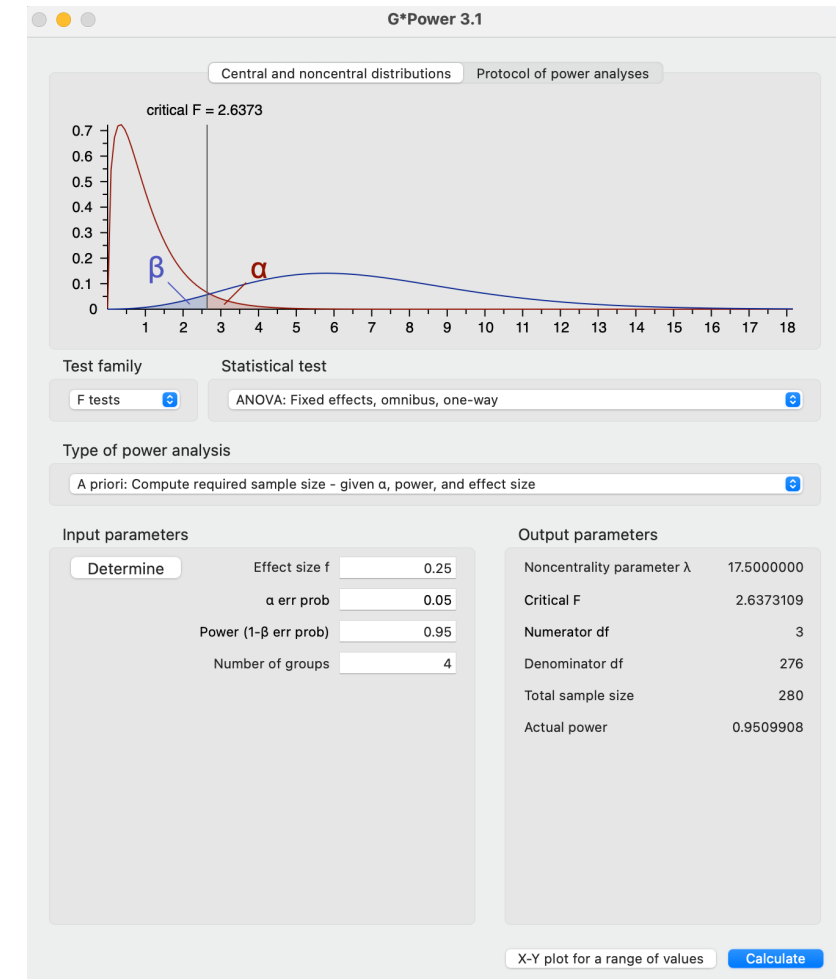
- Describe the sample size of your study. How many units will be analyzed in the study? This could be the number of people, birds, classrooms, plots, interactions, or countries included. If the units are not individuals, then describe the size requirements for each unit. If you are using a clustered or multilevel design, how many units are you collecting at each level of the analysis?
- Example:** Our target sample size is 280 participants. We will attempt to recruit up to 320, assuming that not all will complete the total task.
- More information:** For some studies, this will simply be the number of samples or the number of clusters. For others, this could be an expected range, a minimum, or a maximum.

Sample size rationale

- This could include a power analysis or an arbitrary constraint such as time, money, or personnel.
- Example:** We used the G*Power software to conduct a power analysis. Our goal was to obtain .95 power to detect a medium effect size of .25 at the standard .05 alpha error probability.
- More information:** This gives you the opportunity to specify how the sample size will be determined. A wide range of possible answers is acceptable; remember that transparency is more important than principled justifications. If you state any reason for a sample size upfront, it is better than stating no reason and leaving the reader to “fill in the blanks.” Acceptable rationales include power analysis, an arbitrary number of subjects, or a number based on time or monetary constraints.

Stopping rule

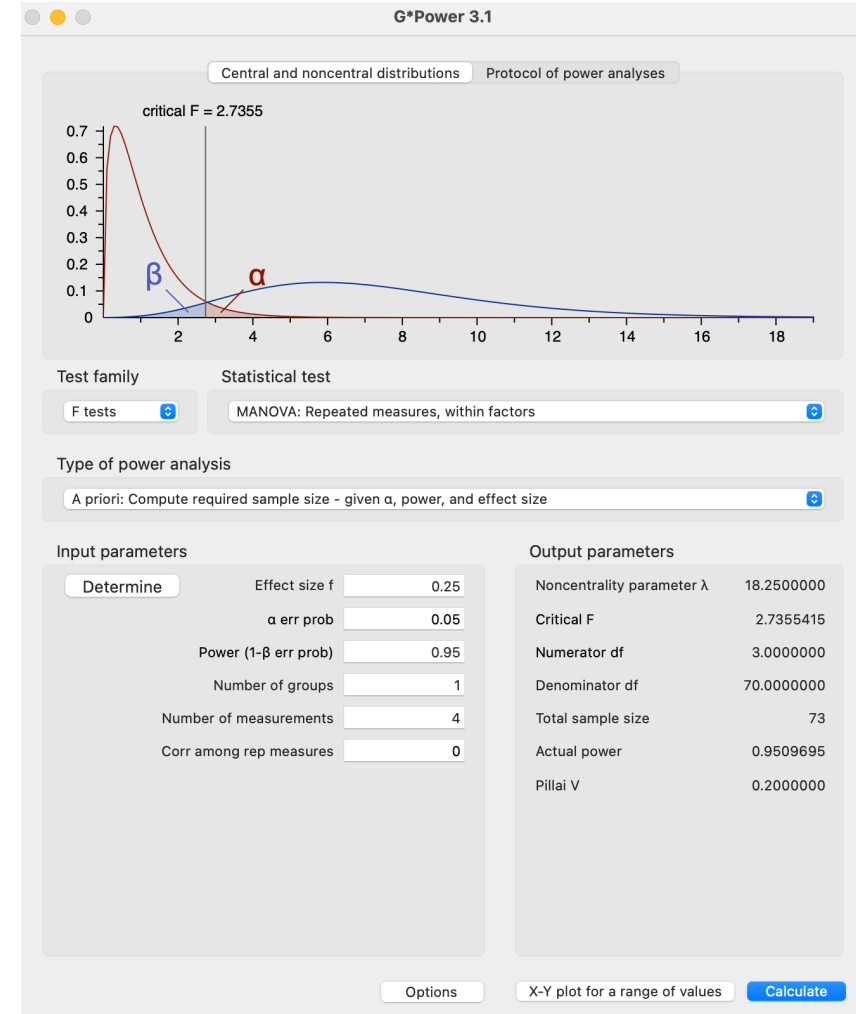
- If your data collection procedures do not give you full control over your exact sample size, specify how you will decide when to terminate your data collection.
- Example:** We will post participant sign-up slots by week on the preceding Friday night, with 20 spots posted per week. We will post 20 new slots each week if, on that Friday night, we have fewer than 320 participants.
- More information:** You may specify a stopping rule based on p-values only in the specific case of sequential analyses with pre-specified checkpoints, alpha levels, and stopping rules. Unacceptable rationales include stopping based on p-values if checkpoints and stopping rules are not specified. If you control your sample size, including a stopping rule is not necessary, though it must be clear in this question or a previous question how the exact sample size is determined.



The case of repeated measures

Between: 280 participants

Within: 73 participants



Effect size

- An **effect size** is **a quantitative measure of the magnitude of a phenomenon** or the strength of a relationship between variables in a study.
- While **a p-value tells you if an effect *exists*, the effect size tells you how *large* or practical that effect actually is** in the real world.
- They are complementary; **you cannot interpret effect sizes unless the results are significant.**

Example: is there a yellow ball in this bag?

Cohen's d

https://en.wikipedia.org/wiki/Effect_size

Sullivan & Fein. 2012. Using Effect Size—or Why the P Value Is Not Enough. . 2012 Sep;4(3):279–282. doi: 10.4300/JGME-D-12-00156.1

Cohen's *d* [[edit](#)]

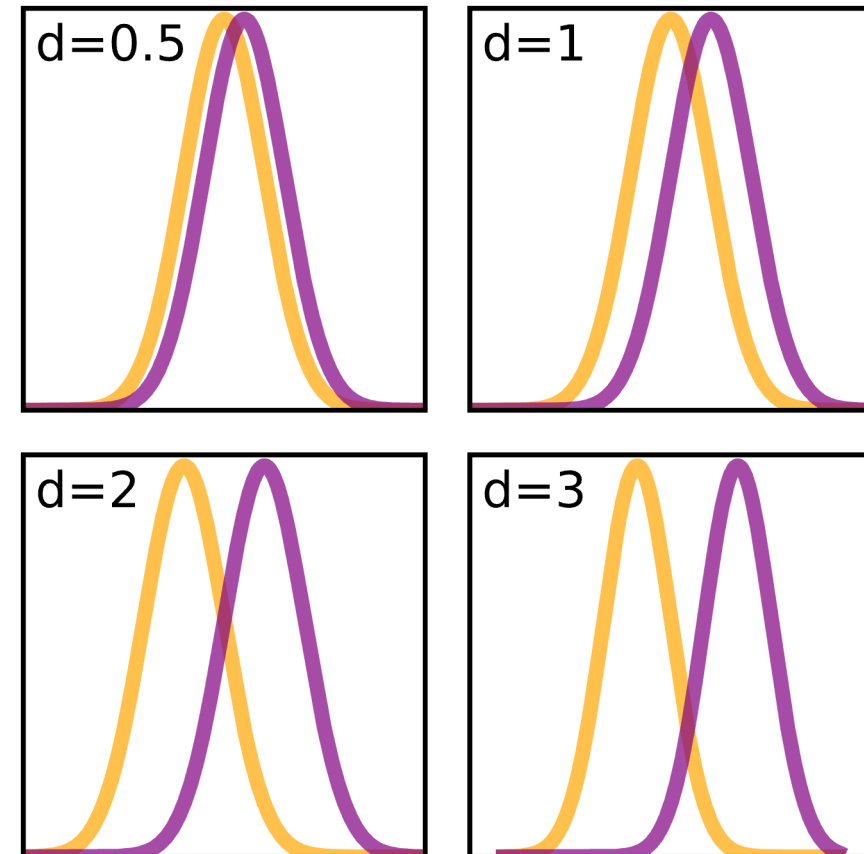
Cohen's *d* is defined as the difference between two means divided by a standard deviation for the data, i.e.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s}.$$

Small: $d = .02$

Medium: $d = .50$

Large: $d = .80$



Statistical significance is the least interesting thing about the results. You should describe the results in terms of measures of magnitude – not just, does a treatment affect people, but how much does it affect them.
-Gene V. Glass

The primary product of a research inquiry is one or more measures of effect size, not P values.
-Jacob Cohen

Sullivan & Fein. 2012. Using Effect Size—or Why the P Value Is Not Enough. . 2012 Sep;4(3):279–282. doi: 10.4300/JGME-D-12-00156.1

Questions?

Contents

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- Why do CS need experiments with people?
- Reproducibility crisis and Open Science
- Experiment protocols and Experiment design
- Power analysis
- **Data analysis using R**
- Ethical approval

Analysis

Analysis plan

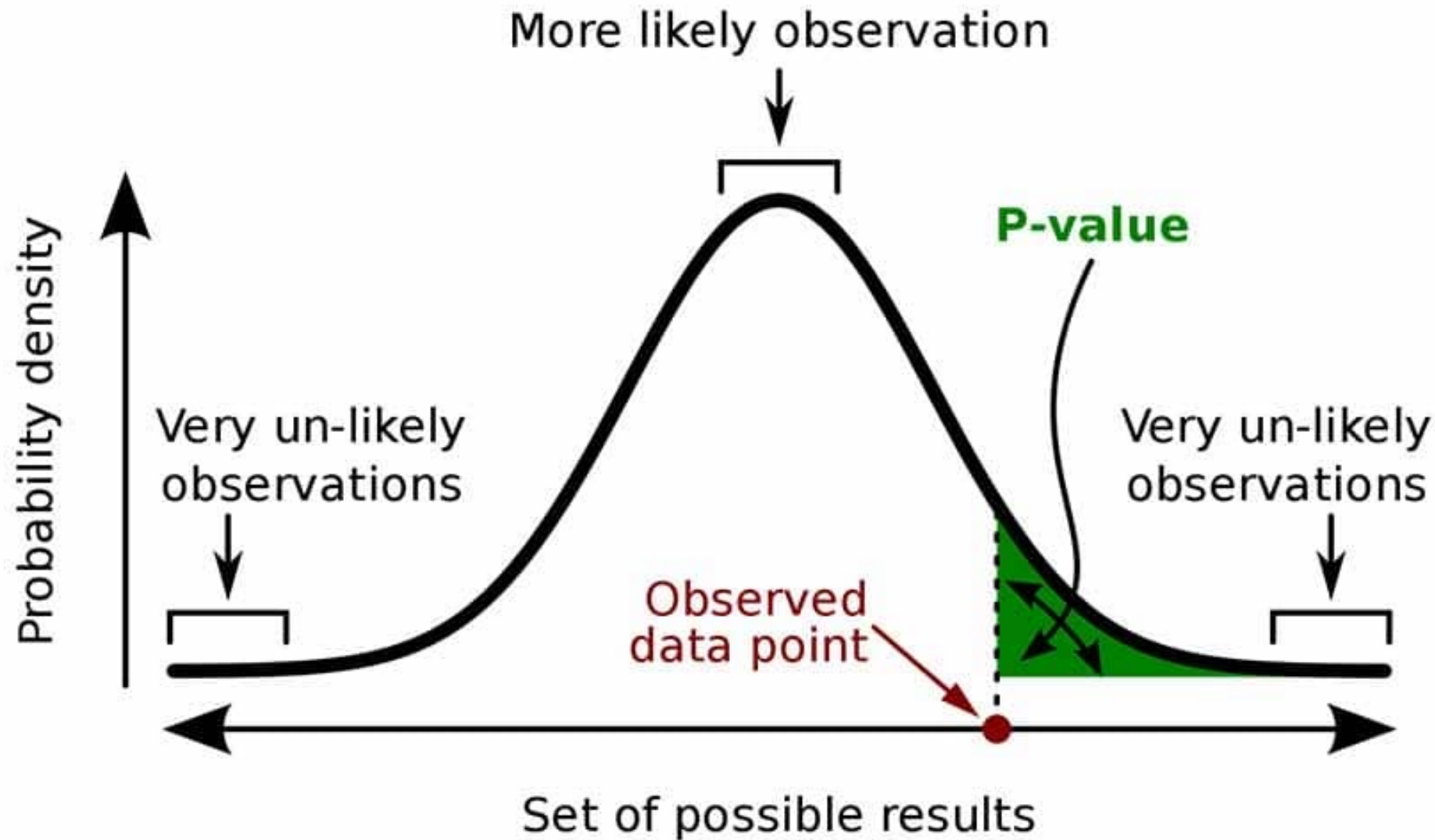
The rule of thumb is that if you don't know how to analyze the data, maybe it is better not to collect it. An exception is if you are creating a database for another person to analyze.

Please remember that all analyses specified below must be reported in the final article, and any additional analyses must be noted as exploratory or hypothesis-generating.

A confirmatory analysis plan must state upfront which variables are predictors (independent) and which are the outcomes (dependent), otherwise, it is an exploratory analysis.

Statistical models

- What statistical model will you use to test each hypothesis? Please include the type of model (e.g., ANOVA, multiple regression, SEM, etc) and the specification of the model (this includes each variable that will be included as predictors, outcomes, or covariates). Please specify any interactions, subgroup analyses, pairwise or complex contrasts, or follow-up tests from omnibus tests. If you plan on using any positive controls, negative controls, or manipulation checks, you may mention that here. Remember that any test not included here must be noted as an exploratory test in your final article.
- **Example:** We will use a one-way between-subjects ANOVA to analyze our results. The manipulated, categorical independent variable is 'sugar,' whereas the dependent variable is our taste index.
- **More information:** This is perhaps the most important and most complicated question within the preregistration. As with all of the other questions, the key is to provide a specific recipe for analyzing the collected data. Ask yourself: Is enough detail provided to run the same analysis again with the information provided by the user? Be aware of instances where the statistical models appear specific but actually leave room for a precise test. See the following examples:
 - If someone specifies a 2x3 ANOVA with both factors within-subjects, there is still flexibility with the various types of ANOVAs that could be run. Either a repeated measures ANOVA (RMANOVA) or a multivariate ANOVA (MANOVA) could be used for that design, which are two different tests.
- If you are going to perform a sequential analysis and check after 50, 100, and 150 samples, you must also specify the p-values you'll test against at those three points. The sequential analysis should be approached with caution, as it may violate the assumptions of traditional hypothesis testing (Neyman-Pearson style) (i.e., violating the duality of the test by considering the next analysis from the results of the analysis).



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

Why R?

Many useful libraries, and it is better documented than Python for scientific purposes



R packages for data science

The tidyverse is an opinionated **collection of R packages** designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

Install the complete tidyverse with:

```
install.packages("tidyverse")
```

Undergraduate Series



Author: Dale Barr

Authors: Lisa DeBruine, Dale Barr
Contact: Lisa DeBruine
Contributors: Rebecca Lai

井関龍太のページ



ANOVA君とは†

ANOVA君のファイル十

【リリースノート】
2023-05-01 anovakun version 4.8.9公開

<https://riseki.cloudfree.jp/?ANOVA%E5%90%9B>

Questions?

Contents

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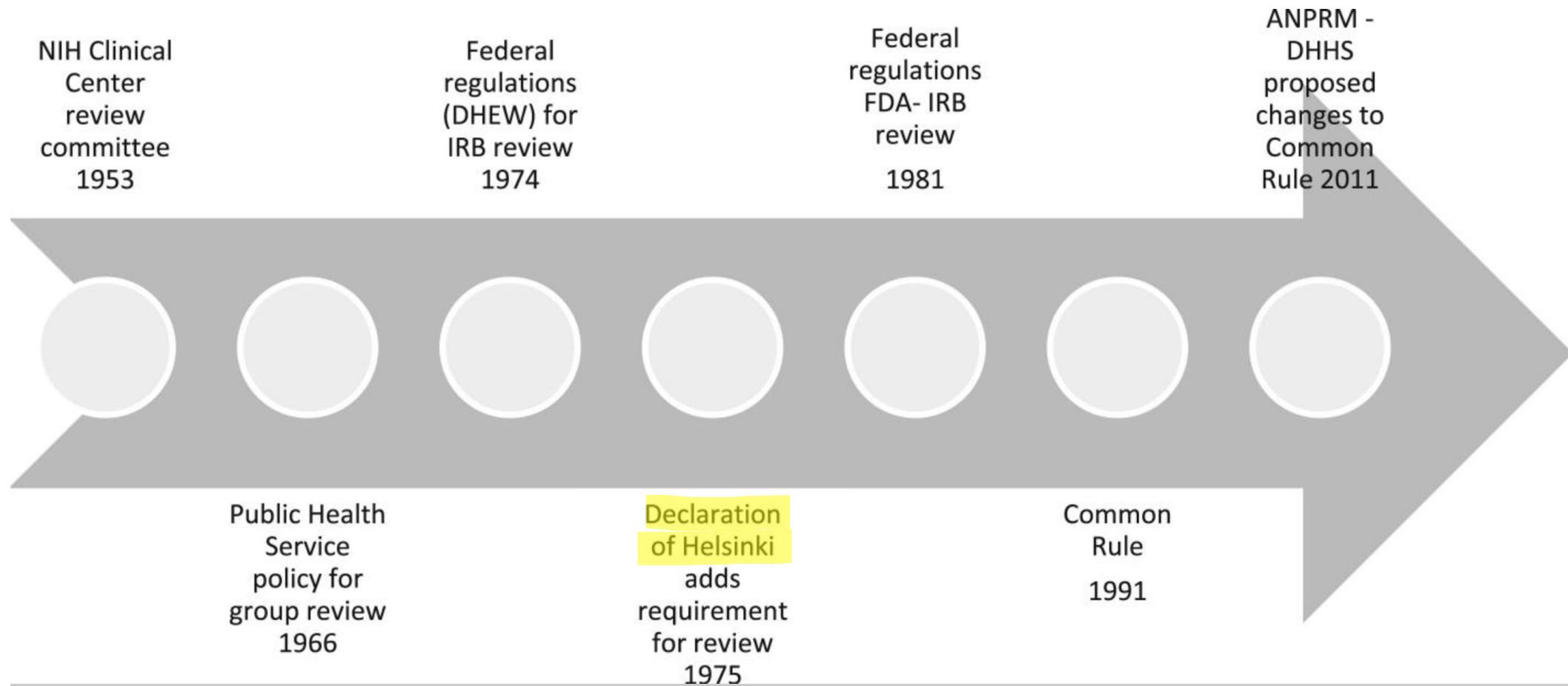
- Why do CS need experiments with people?
- Reproducibility crisis and Open Science
- Experiment protocols and Experiment design
- Power analysis
- Data analysis using R
- **Ethical approval**

Institutional Review Board (IRB)

- An independent committee of scientists, ethicists, and community members
- It reviews and monitors research involving human subjects to ensure their rights, safety, and welfare are protected.
- They ensure:
 - Participant protection
 - Ethical standard enforcement
 - Scientific integrity

Grady. 2015. Institutional Review Boards, Purpose and Challenges.
Chest148(5):1148–1155. doi: 10.1378/chest.15-0706

Institutional Review Board (IRB)



Grady. 2015. Institutional Review Boards, Purpose and Challenges. Chest148(5):1148–1155. doi: 10.1378/chest.15-0706

The IRB decides whether you can conduct your experiment

The procedure protects participants, and also protects you against unexpected outcomes

Good practices and ethics

1. Guard human participants, their dignity, rights and welfare
2. Protocol for recruitment and explanation of the study
3. Informed consent
4. Availability of data and code (data transparency)
5. Ethics committee board approval
6. Be sensitive towards the individuals that participate in your study
7. Make clear that their personal information is safe

Questions?

Biosignal Processing for Affective Computing Applications in **Virtual Reality**

Monica Perusquía-Hernández

Nara Institute of Science and Technology (NAIST)

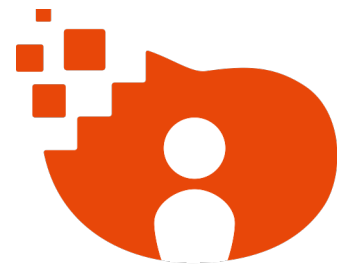
2026.09.07



Topics

Time	Topic
9:00 – 9:30	Introduction to the tutorial • Virtual reality as wearable technology that alters and expands reality to create (affective) experiences • Biosignals measurement as wearable technology to assess affective states • Motivation to use both VR and biosignals to study emotion
9:30 – 9:50	Video-based demo of data collection
9:50 – 10:30	Biosignal processing theory • Introduction to example biosignals • Practical recommendations when measuring biosignals in general • Synchronization with other devices
11:00 – 11:30	Biosignal processing practice • Hands-on signal processing for each signal
11:30 – 12:00	Practical tips to integrate biosignal measurement and Virtual Reality • Tips to port experiments to a headset • Synchronization with Unity • Hardware • Software • Mitigation of artifacts when working with a HMD

Thank you very much



Cybernetics &
Reality Engineering